آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Mathematics Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of the Higher Secondary School Certificate (HSSC) Part II Mathematics. It contains comments on candidates' responses to the 2025 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfill the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs, which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why', or 'what' may also be used. It is imperative to refer to the command word guide available on AKU-EB website for understanding the expectations of the command word.

General Observations

Overall, most candidates performed well in questions involving the **derivative of exponential and logarithmic functions, partial integration, straight-line equations and altitudes, the hyperbola and its components, and properties of perpendicular vectors**. These areas showed that candidates had a generally sound grasp of the key concepts and could apply them effectively.

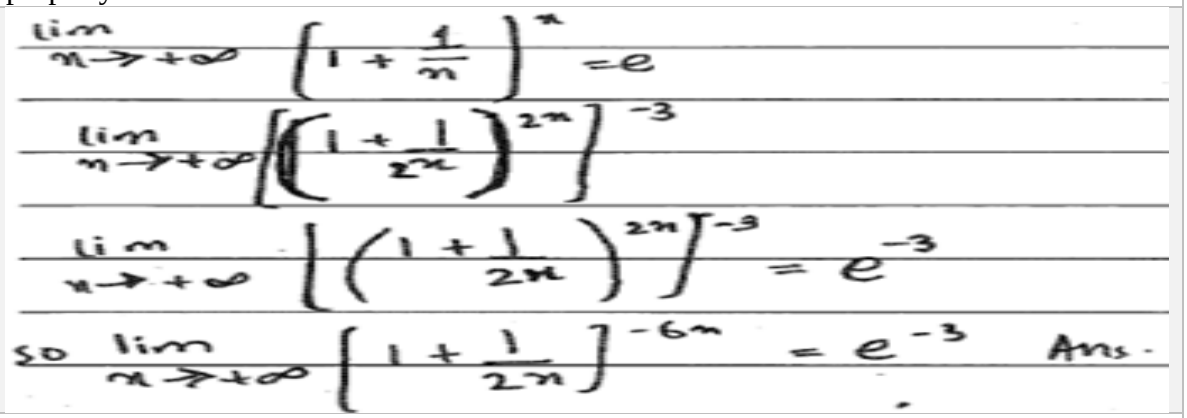
However, there are several topics where candidates struggled and would benefit from more focused instruction and practice. These include:

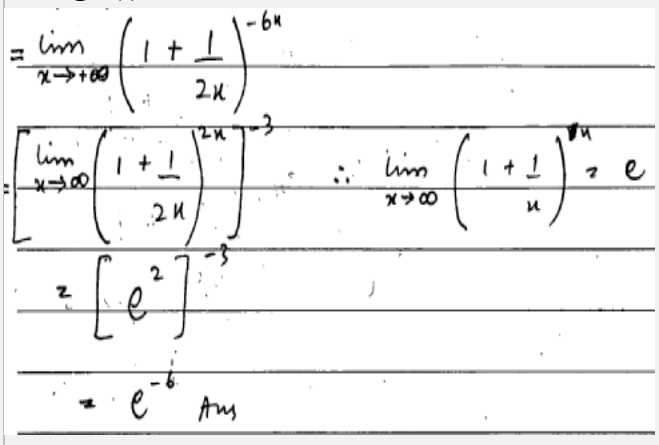
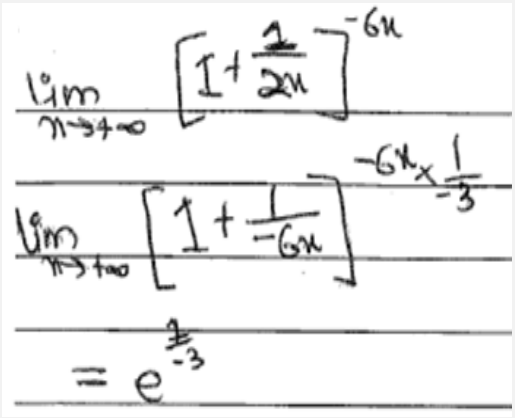
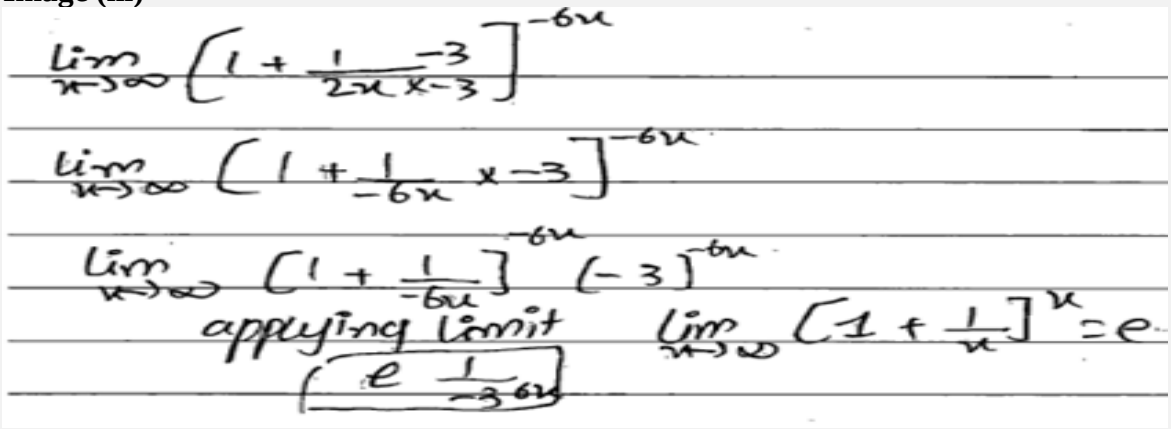
- **Limits of functions**
- **Differentiation using first principles**
- **Inverse hyperbolic functions**
- **Tangents to curves**
- **Higher-order derivatives**
- **Integration by substitution**
- **Feasible region in linear programming**
- **Circle and its key properties**
- **Parabola**
- **Ellipse and its components**
- **Rotation of axes**
- **Cross product of two vectors**

Teachers are encouraged to place greater emphasis on these topics through guided teaching and regular practice to help improve understanding and application.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

Question No. 1	
Question Text	Evaluate the limit of the given function. $\lim\limits_{x\to+\infty}\left(1+\frac{1}{2x}\right)^{-6x}$
SLO No.	12.9.1
SLO Text	Evaluate the limits of functions of the following forms: a. $\frac{x^n-a^n}{x-a}, \frac{x-a}{\sqrt{x}-\sqrt{a}}$ when $(x\to a)$ b. $\left(1+\frac{1}{x}\right)^x$ when $(x\to\infty)$ c. $(1+x)^{\frac{1}{x}}, \frac{\sqrt{x+a}-\sqrt{a}}{x}, \frac{a^x-1}{x}, \frac{(1+x)^n-1}{x}$, and $\frac{\sin x}{x}$ when $(x\to 0)$;
Max Marks	3
Cognitive Level	*A
Checking Hints	1 mark for writing splitting -6*x* 1 mark for making the limit feasible for applying the formula 1 mark for evaluating the limit as a power of e
Overall Performance	Overall, most candidates found this question difficult. Only a few demonstrated a clear understanding of evaluating limits. While some applied the correct methods and used algebraic reasoning effectively, many struggled with splitting exponents or applying the correct limit rules.
Description of Better Responses	Candidates who performed well approached the question methodically. They recognised the expression as a standard exponential limit form and correctly used the identity: $\lim\limits_{x\to\infty}\left(1+\frac{1}{nx}\right)^{nx}=e$, In this case, they identified that the given expression and successfully applied: $\lim\limits_{x\to\infty}\left(\left(1+\frac{1}{2x}\right)^{2x}\right)^{-3}=e^{-3}$. This showed a solid understanding of exponential limits and laws of indices. Their work was clear, correctly simplified, and used limit identities properly.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* often showed confusion about the structure of the expression. Many candidates attempted direct substitution or used incorrect limit rules. Some confused this expression with polynomial or rational limits, rather than recognising the exponential form. Others incorrectly applied logarithmic steps or attempted to expand the expression without

	understanding the context. Many responses lacked essential steps, skipped justifications, or provided final answers without working, resulting in lost marks. In addition, several candidates did not appear familiar with the form $\left(1+\frac{1}{x}\right)^{x} \to e$, and as a result, could not relate the given expression to the standard exponential limit.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Prepare cards: some with limits in various forms, others with their simplified exponential results.
- Students work in pairs to match each limit with its value.
- Give a complex limit.
- First student substitutes ($t = 2x$), passes to the next, who applies the standard limit rule, next simplifies to the final answer.
- Teams race to complete.
- Give students similar limits with different constants.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2a Candidates were given the choice to attempt any TWO out of the three questions: 2a, 2b and 2c.	
Question Text	Using differentiation by first principle, differentiate $y = \sin(1-2x)$.
SLO No.	13.4.1
SLO Text	Find the derivative of trigonometric functions by first principles.
Max Marks	5
Cognitive Level	A
Checking Hints	1 mark for taking expressing as $y + \delta y = \sin(1 - 2(x + \delta x))$ 1 mark for writing as difference as $\delta y = \sin(1-2(x+\delta x)) - \sin(1-2x)$ 1 mark for dividing by as δx on both sides 1 mark for applying limit 1 mark for evaluating limit correctly
Overall Performance	Overall, a few of the candidates attempted this question. While some demonstrated a basic understanding of the requirement, only a small number successfully applied the method of differentiation using first principles. This suggests that although the concept may have been taught, students need more targeted instruction and practice to progress from understanding the theory to applying it accurately.
Description of Better Responses	*Better responses* showed a clear understanding of the method. Candidates began by expressing the change in function values using a small increase in the input. They substituted correctly and simplified the resulting expression using relevant trigonometric identities. The difference was then divided by the increment, and the limit was carefully evaluated. Each step was logically followed, with clear working and accurate reasoning, leading to the correct derivative.
Images of Better Responses	**Image (i)**

$$\frac{f(x+\delta x)-f(x)}{\delta x} = \frac{dy}{dx}$$

$$\therefore f(x+\delta x) - f(x) = \sin(1-2(x+\delta x)) - \sin(1-2x)$$

$$= \sin(1-2x-2\delta x) - \sin(1-2x)$$

$$\because \sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$= 2\cos\left(\frac{1-2x-2\delta x+1-2x}{2}\right)\sin\left(\frac{-2x-2\delta x-1+2x}{2}\right)$$

$$= 2\cos\left(\frac{1-2x-2\delta x+1-2x}{2}\right)\sin(-\delta x)$$

Dividing both sides by δx and Multiply and divide (-1) on R.H.S

$$\frac{f(x+\delta x)-f(x)}{\delta x} = -2\cos\left(\frac{1-2x-2\delta x+1-2x}{2}\right)\frac{\sin(-\delta x)}{-\delta x}$$

Apply $\lim_{\delta x \to 0}$, we have $\frac{dy}{dx} = -2\cos\left(\frac{2-4x-2(0)}{2}\right)(1)$

$\therefore \left(\lim_{\delta x \to 0}\frac{\sin \delta x}{\delta x} = 1\right)$

$$dy/dx = -2\cos(1-2x)$$

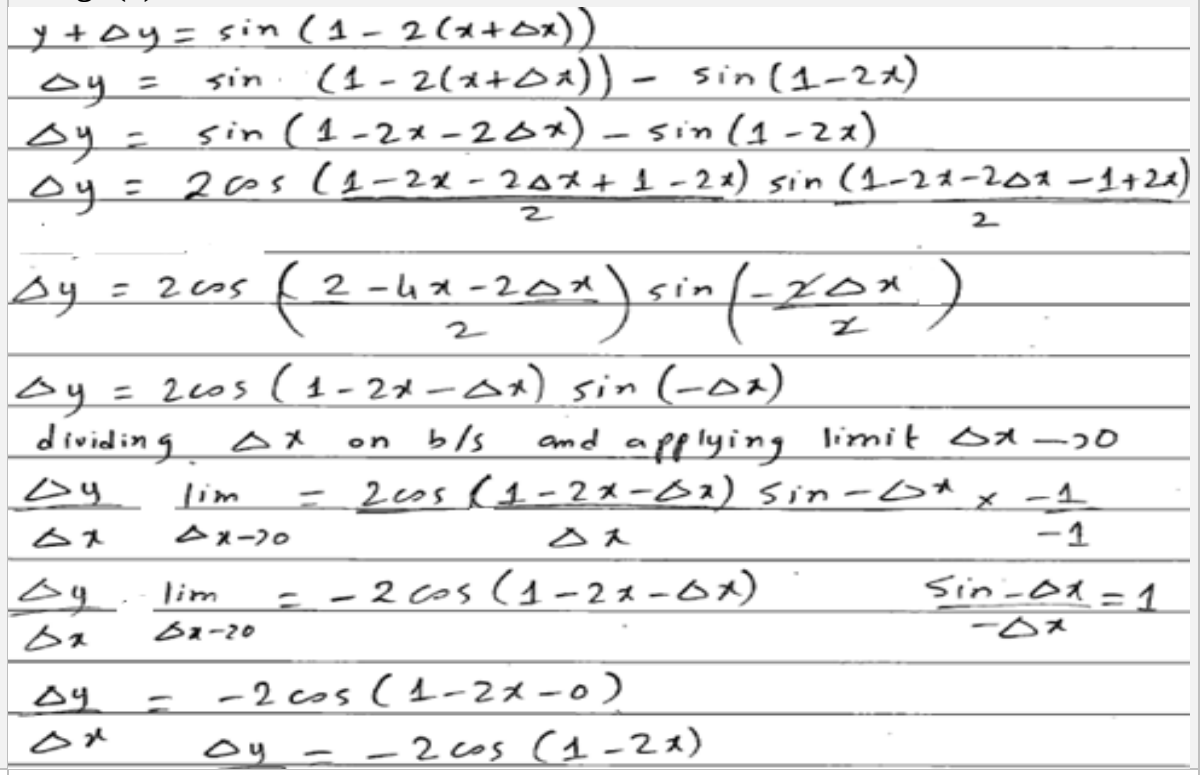
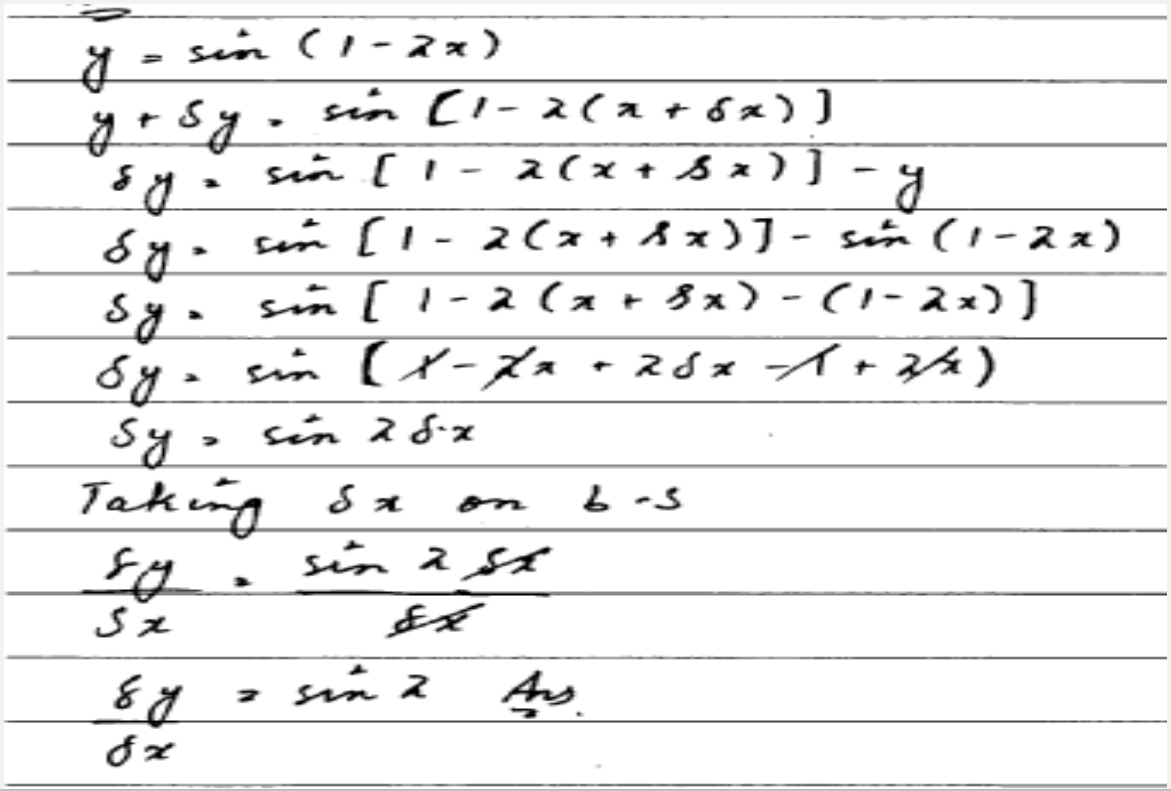
	Image (ii)  $y + \Delta y = \sin(1 - 2(x+\Delta x))$ $\Delta y = \sin(1 - 2(x+\Delta x)) - \sin(1-2x)$ $\Delta y = \sin(1 - 2x - 2\Delta x) - \sin(1-2x)$ $\Delta y = 2\cos\left(\frac{1-2x-2\Delta x+1-2x}{2}\right)\sin\left(\frac{1-2x-2\Delta x-1+2x}{2}\right)$ $\Delta y = 2\cos\left(\frac{2-4x-2\Delta x}{2}\right)\sin\left(\frac{-2\Delta x}{2}\right)$ $\Delta y = 2\cos(1-2x-\Delta x)\sin(-\Delta x)$ dividing Δx on b/s and applying limit $\Delta x \to 0$ $\frac{\Delta y}{\Delta x} \lim_{\Delta x \to 0} = \frac{2\cos(1-2x-\Delta x)\sin-\Delta x}{\Delta x} \times \frac{-1}{-1}$ $\frac{\Delta y}{\Delta x} \lim_{\Delta x \to 0} = -2\cos(1-2x-\Delta x)$ $\frac{\sin-\Delta x}{-\Delta x} = 1$ $\frac{\Delta y}{\Delta x} = -2\cos(1-2x-0)$ $\frac{\Delta y}{\Delta x} \quad \Delta y = -2\cos(1-2x)$
Description of Weaker Responses	*Weaker responses* often included vague or incorrect expressions. Many candidates attempted to differentiate directly using known derivatives of sine functions, which did not meet the question's demand for using the first principle. Others wrote down the first principle formula but struggled to apply it to the given function, often failing to correctly expand or simplify the trigonometric expressions. There were also frequent errors in using or applying trigonometric identities, such as misusing the sine subtraction identity or failing to recognise the inner function $1-2x$. In several responses, candidates skipped essential working steps, resulting in unclear or incomplete solutions that could not be awarded full marks.
Images of Weaker Response	**Image (i)**  $y = \sin(1-2x)$ $y + \delta y = \sin[1-2(x+\delta x)]$ $\delta y = \sin[1-2(x+\delta x)] - y$ $\delta y = \sin[1-2(x+\delta x)] - \sin(1-2x)$ $\delta y = \sin[1-2(x+\delta x) - (1-2x)]$ $\delta y = \sin[1-2x+2\delta x-1+2x)$ $\delta y = \sin 2\delta x$ Taking δx on b-s $\frac{\delta y}{\delta x} = \frac{\sin 2\delta x}{\delta x}$ $\frac{\delta y}{\delta x} = \sin 2$ Ans.

Image (ii)

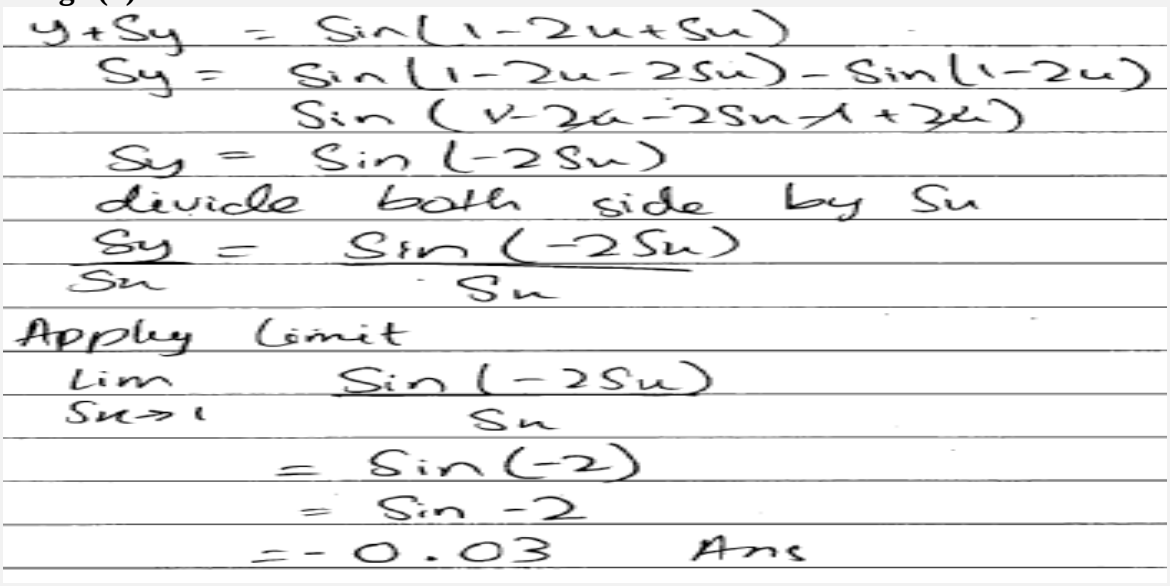
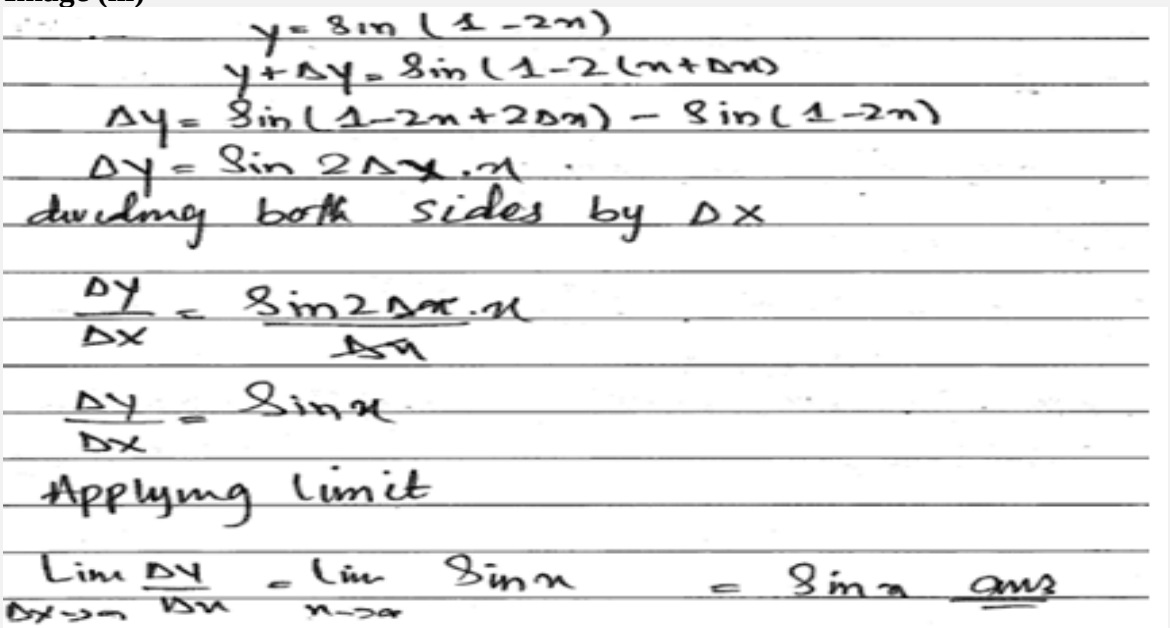


Image (iii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts)	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

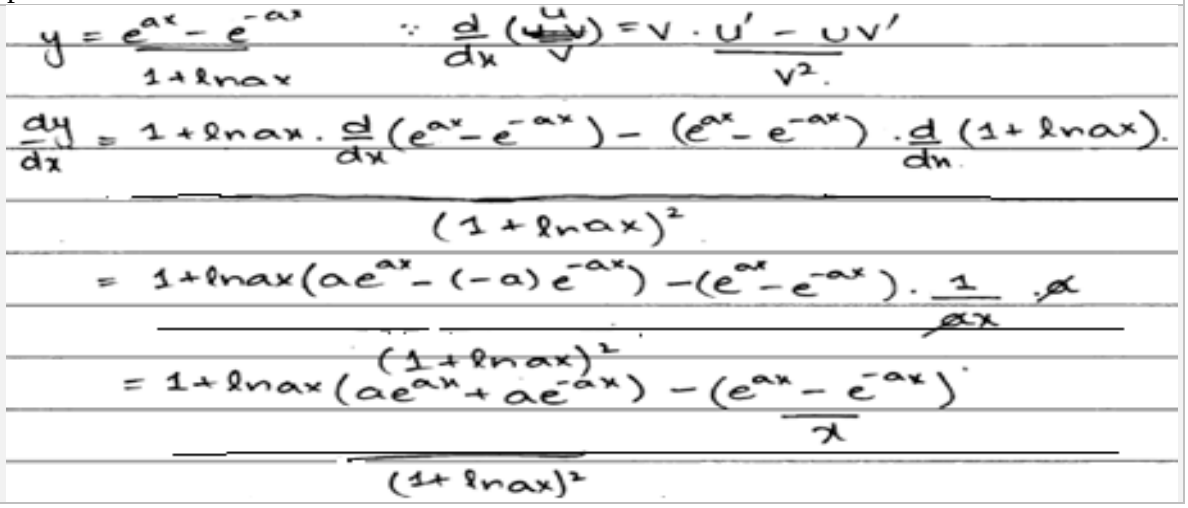
• Review past paper questions on the concept • Utilise the resource guide for additional materials		

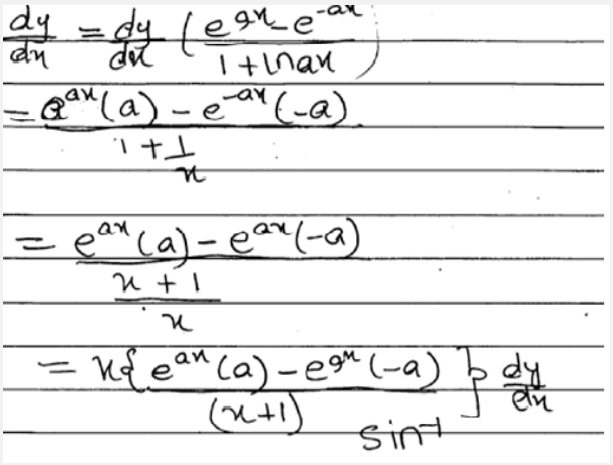
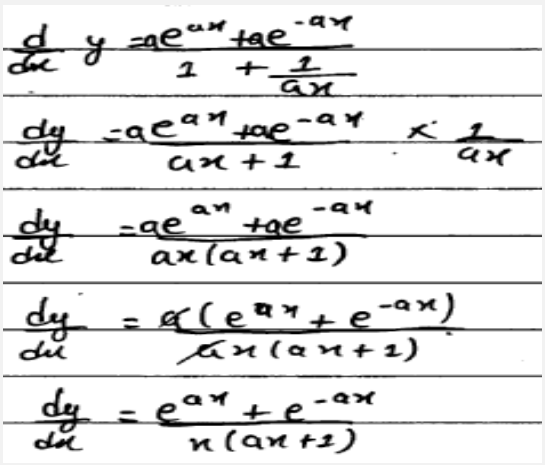
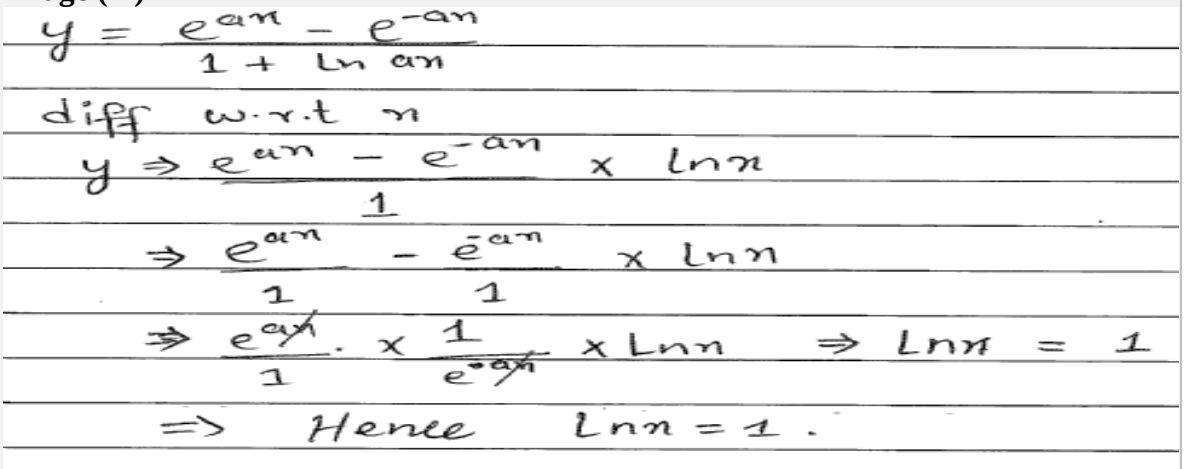
Any Additional Suggestion:

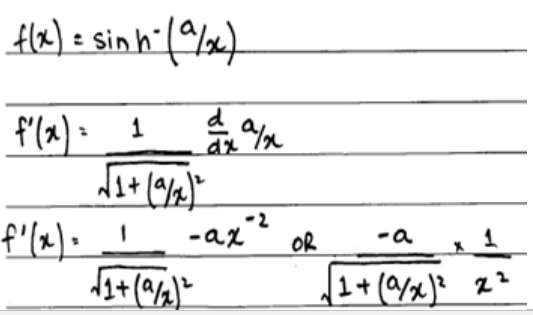
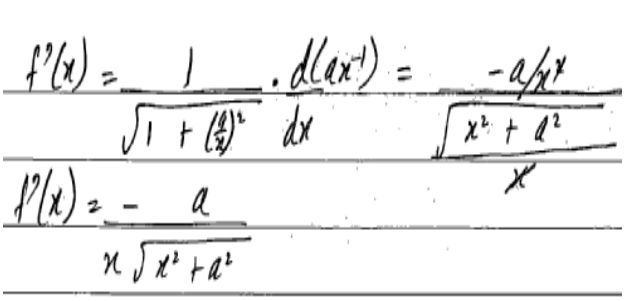
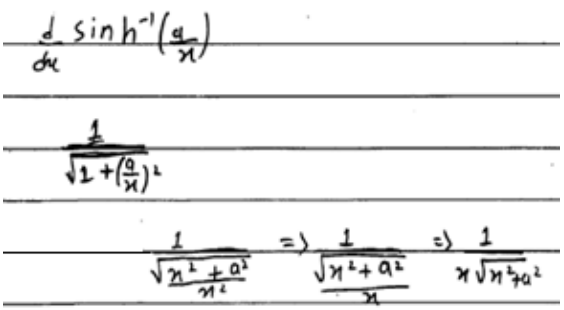
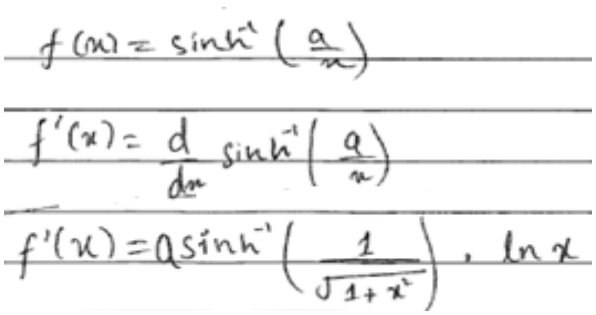
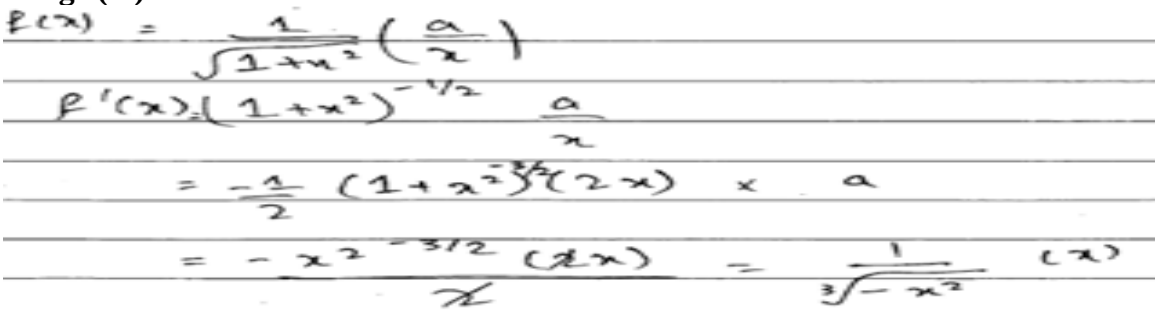
- Students use an online graphing tool like Desmos or GeoGebra to plot sine or cosine curves and estimate slopes at various points, then compare with derivative values from first principles.
- In pairs, students use a spreadsheet or calculator to compute slope values for smaller increments of h and observe the pattern approaching the derivative.
- Provide incomplete trigonometric identities for students to complete and then apply in the derivation.

Question No. 2b

Candidates were given the choice to attempt any TWO out of the three questions: 2a, 2b and 2c.

Question Text	i. If $y = \dfrac{e^{ax} - e^{-ax}}{1 + \ln ax}$, then find $\dfrac{dy}{dx}$.
SLO No.	13.5.3
SLO Text	Find the derivative of exponential and logarithmic functions by using direct method.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for applying quotient rule 1 mark for correctly differentiating $e^{ax} - e^{-ax}$ 1 mark for correctly differentiating $1 + \ln ax$
Overall Performance	Overall, most of the candidates attempted this question. Some showed a fair understanding of the rules for differentiating exponential and logarithmic functions. However, many found it difficult to apply these rules correctly when dealing with a quotient. This highlights the need for improved instruction on combining various differentiation methods.
Description of Better Responses	*Better responses* showed a solid grasp of the quotient rule and the differentiation of both exponential and logarithmic functions. Candidates began by identifying the numerator and denominator correctly and proceeded to apply the quotient rule in a structured manner. They differentiated the exponential expression accurately, handled the logarithmic function appropriately, and simplified the resulting expression with care. Each step was completed logically, leading to a correct and simplified derivative. These responses showed strong algebraic manipulation skills and clear understanding of function properties throughout the process.
Image of Better Response	

Description of Weaker Responses	*Weaker responses* revealed several gaps in understanding. Many candidates struggled to recall or apply the quotient rule correctly. Others incorrectly differentiated the exponential or logarithmic parts, sometimes confusing the rules for each. In some cases, the quotient rule was misapplied or replaced with unrelated methods, leading to incorrect working and final answers. A number of responses lacked simplification or contained algebraic errors, indicating limited familiarity with the necessary techniques.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 
Question Text	ii. If $f(x)=\sinh^{-1}\left(\frac{a}{x}\right)$, then find $f'(x)$. (**Note:** $\frac{d}{dx}\sinh^{-1}x=\frac{1}{\sqrt{1+x^2}}$)
SLO No.	13.6.1(b)
SLO Text	Find the derivative of: a. hyperbolic functions (sinh *x*, cosh *x*, tanh *x*, cosech *x*, sech *x* and coth *x*), b. inverse hyperbolic functions $\sinh^{-1}x, \cosh^{-1}x, \tanh^{-1}x, \text{cosech}^{-1}x, \text{sech}^{-1}x$, and $\coth^{-1}x$).
Max Marks	2
Cognitive Level	A
Checking Hints	1 mark for applying the formula of *sin*h^{-1} 1 mark for differentiating the angle
Overall Performance	Overall, a few of the candidates showed a fair understanding of differentiating inverse hyperbolic functions. However, many found it difficult to apply the correct rules, especially when the function involved a variable expression in terms of *x*. This suggests that more practice is needed to help candidates handle such problems with confidence.

Description of Better Responses	*Better responses* demonstrated a clear understanding of both inverse hyperbolic functions and angle differentiation. Candidates correctly identified the need to apply the chain rule, first differentiating the inverse hyperbolic function, and then multiplying by the derivative of the angle a/x. Their work was accurate, step-by-step, and well-structured, showing confidence in handling both the standard form and the composite function. These responses reflected good understanding and accurate simplification.
Images of Better Responses	**Image (i)**  **Image (ii)** 
Description of Weaker Responses	*Weaker responses* showed confusion between hyperbolic and trigonometric functions, with some candidates attempting to use incorrect identities or formulas. A common mistake was differentiating $\sinh^{-1}(x)$ directly without considering the angle being a/x, thus neglecting the chain rule. Others either left the response incomplete or used incorrect forms of the derivative, resulting in wrong answers. Several candidates also showed uncertainty in simplifying their results, further reducing accuracy.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:

- Students use Desmos or GeoGebra to plot exponential, logarithmic, hyperbolic, and quotient-type functions along with their derivatives, matching each function to its derivative.
- They then solve a mix of problems covering all four types, applying the correct rules.
- Given derivative expressions, they guess the original functions, and finally create a real-life model (such as population growth, sound intensity, cable shapes, or efficiency ratios), differentiate it, plot both the original and derivative, and explain the relationship.

Question No. 2c

Candidates were given the choice to attempt any TWO out of the three questions: 2a, 2b and 2c.

Question Text	A function is defined as $y = \frac{1}{3}x^3 - x^2 - 3x + 5$. Find the value(s) of x at which the i. tangent is parallel to x-axis. ii. function obtains the relative highest value.
SLO No.	14.3.4 and 14.3.6
SLO Text	Find the point on a curve where the tangent is parallel to the given line. Apply second derivative rule to examine a given function for extreme values.
Max Marks	5
Cognitive Level	A
Checking Hints	i. 1 mark for finding the first derivative 1 mark for putting derivative zero 1 mark for finding the values of x ii. 1 mark for second derivative 1 mark for applying the second derivative rule
Overall Performance	Overall, a smaller number of candidates attempted this question. While some demonstrated a good understanding of using differentiation to analyse the behaviour of functions, many struggled to apply the process accurately. A common issue was the incorrect use of factorisation to find critical points after differentiating.
Description of Better Responses	*Better responses* demonstrated a correct approach to solving both parts of the question. Candidates accurately differentiated the given function and equated the first derivative to zero to find critical points. They correctly interpreted that a tangent parallel to the x-axis occurs where the derivative is zero, and identified these values of x through proper factorisation and solving. In the second part, these candidates used appropriate reasoning to determine which of the critical points corresponded to a relative maximum. Some supported their answers with the second derivative test or by evaluating the nature of the function around the critical points. Their solutions were clearly structured and included both the correct value(s) of x and the corresponding maximum value of the function.

Image of Better Response	$\frac{dy}{dx} = x^2 - 2x - 3$ — slope of tangent Slope of x-axis = 0 and since tangent is parallel to x-axis so slope of tangent = 0 $x^2 - 2x - 3 = 0$ $(x-3)(x+1) = 0$ $x = 3$ or $x = -1$ hence the values of x are 3 and −1. $f''(x) = 2x - 2$ $f''(3) = 2(3) - 2 \Rightarrow 4$ $f''(3) > 0$ so it's a minimia. $f''(-1) = 2(-1) - 2 \Rightarrow -4$ $f''(-1) < 0$ so it's a maxima hence function obtains the relative highest value at $x = -1$ and $f(-1) = \frac{1}{3}(-1)^3 - (-1)^2 - 3(-1) + 5 \Rightarrow \frac{20}{3}$
Description of Weaker Responses	*Weaker responses* often showed difficulty in differentiating the cubic expression. Common issues included errors in applying basic differentiation rules or omitting terms altogether. Several candidates reached the correct first derivative but failed to solve it correctly, particularly when factorisation was required. Others did not set the derivative equal to zero, which is essential for identifying where the tangent is horizontal or where maxima and minima occur. In the second part, many responses did not include a method to confirm whether the point found was a maximum or minimum. Some candidates either guessed the value or provided the critical point without evaluating the function at that point. These gaps suggest a limited understanding of how to use differentiation to analyse the behaviour of functions.
Image of Weaker Response	i. tangent is parallel to x-axis. (3 Marks) $y = \frac{1}{3}x^3 - x^2 - 3x + 5 = 0$ Taking L.C.M on b.s $\frac{3x^3 - 3x^2 - 9x + 15}{3} = \frac{0}{3}$ $3x(x^2 - x - 3 + 5) = 0$ $x^2 - x - 3 + 5 = \frac{0}{3x} = 0$ $x + 2 = 0$ $x = -2$ Ans. ii. function obtains the relative highest value. (2 Marks) $y = \frac{1}{3}(-2)^3 - (-2)^2 - 3(-2) + 5 = 0$ $\frac{-8}{3} - 4 + 6 + 5 = 0$ $-\frac{8}{3} + 7 = 0$ $\therefore$ $y = -1$ Taking L.C.M $\frac{-8+21}{3}$

Suggestions for improvement (Highlight all that apply)

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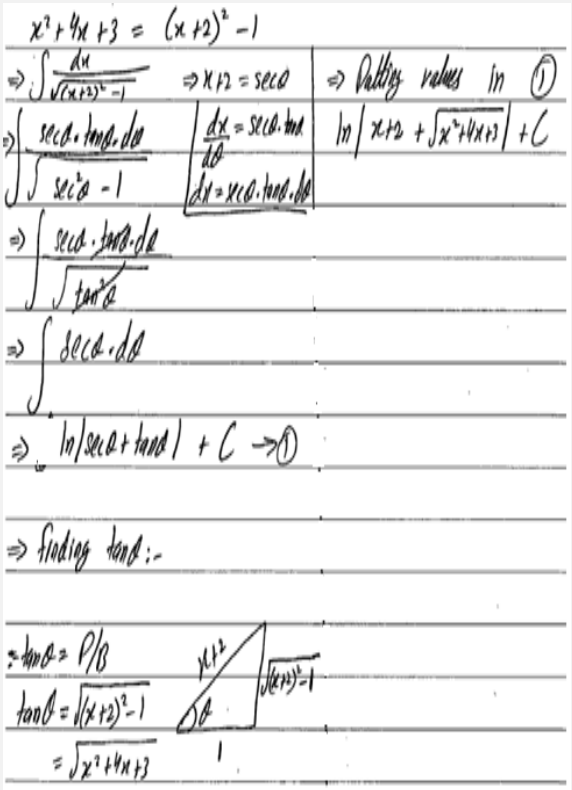
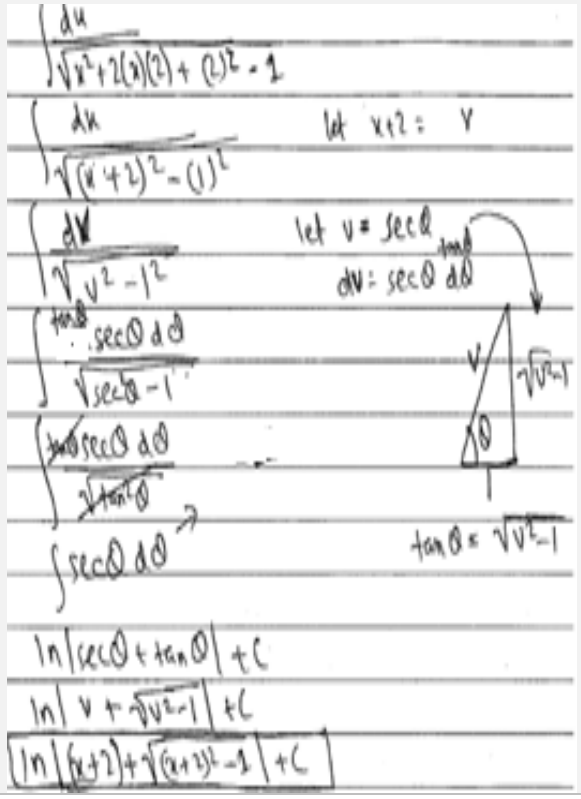
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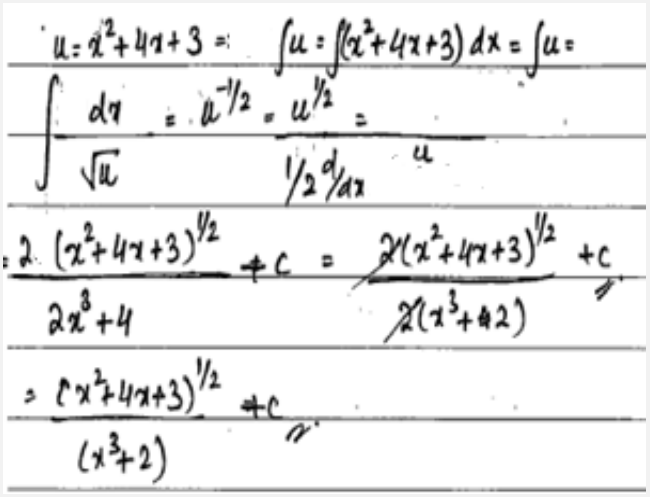
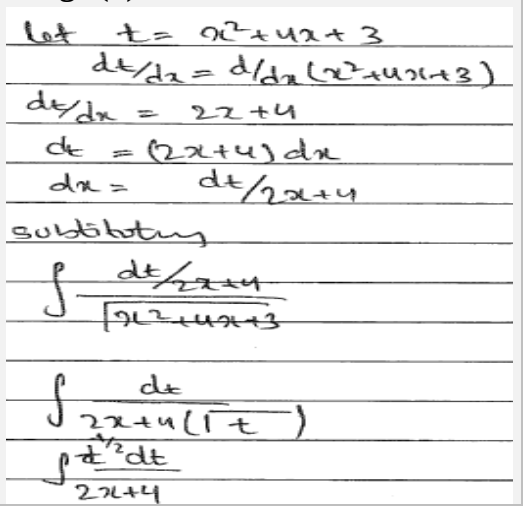
- Students use a graphing tool to plot the curve and given line, visually locate where slopes match, and confirm algebraically.
- Create a card set with different curves and line equations for students to match and find the tangent point.
- In small groups, give each team a different function and line, with the goal of quickly identifying the correct point(s).
- Students plot a function and its first and second derivatives in Desmos to identify maxima, minima, and points of inflection.
- Provide functions with missing slope or concavity details for students to complete using the second derivative test.

Question No. 3a

Candidates were given the choice to attempt any TWO out of the three questions: 3a, 3b and 3c.

Question Text	Using integration by substitution, evaluate $\int \frac{dx}{\sqrt{x^2+4x+3}}$.
SLO No.	16.3.3b
SLO Text	Apply the method of substitution to evaluate the integrals of the following types: a. $\int \frac{dx}{a^2-x^2}$, $\int \sqrt{a^2-x^2}\,dx$, $\int \frac{dx}{\sqrt{a^2-x^2}}$, b. $\int \frac{dx}{a^2+x^2}$, $\int \sqrt{a^2+x^2}\,dx$, $\int \frac{dx}{\sqrt{x^2+a^2}}$, c. $\int \frac{dx}{x^2-a^2}$, $\int \sqrt{x^2-a^2}\,dx$, $\int \frac{dx}{\sqrt{x^2-a^2}}$;
Max Marks	5
Cognitive Level	A
Checking Hints	1 mark for writing in the form $\int \frac{dx}{\sqrt{(x+2)^2-1}}$. 1 mark for substituting as $x+2=\sec t$ 1 mark for converting dx into dt

	1 mark for integrating $\int \sec t\, dt$ and writing as $\ln\left(\sec t + \tan t\right) + c$ 1 mark for re-substituting and writing the integral in the terms of x as $\ln\left\lvert (x+2) + \sqrt{x^2+4x+3}\right\rvert + C$
Overall Performance	Overall, only a small number of candidates attempted this question. Of those who did, a few of them showed a good understanding of integration using trigonometric substitution. However, many struggled with choosing the correct trigonometric substitution, differentiating it properly, and carrying out the re-substitution step.
Description of Better Responses	*Better responses* showed a clear understanding of the method of trigonometric substitution in integration. Candidates first completed the square for the quadratic expression under the square root and then identified the appropriate trigonometric substitution to simplify the integrand. Their differentiation was accurate, and they correctly replaced all parts of the integral with expressions in terms of the new variable. After integrating the simplified form, they resubstituted it back into terms of x accurately, leading to the correct final result. These responses were well-structured and logically sequenced, demonstrating a full grasp of each stage of the process.
Images of Better Responses	**Image (i)**  **Image (ii)** 
Description of Weaker Responses	*Weaker responses* often revealed confusion over the substitution technique. Many candidates selected inappropriate algebraic substitutions, which did not simplify the integral correctly. Others attempted to use trigonometric substitution but made errors in differentiating the substituted variable or did not fully transform the entire integral. In some cases, even when integration was performed correctly, candidates failed to re-substitute back into the original variable or did so incorrectly, resulting in an incomplete or incorrect final answer. These issues highlight a lack of understanding of the full substitution process and its application to integrals involving quadratic expressions.

Images of Weaker Responses	Image (i)	Image (ii)
		

Suggestions for improvement (Highlight all that apply)

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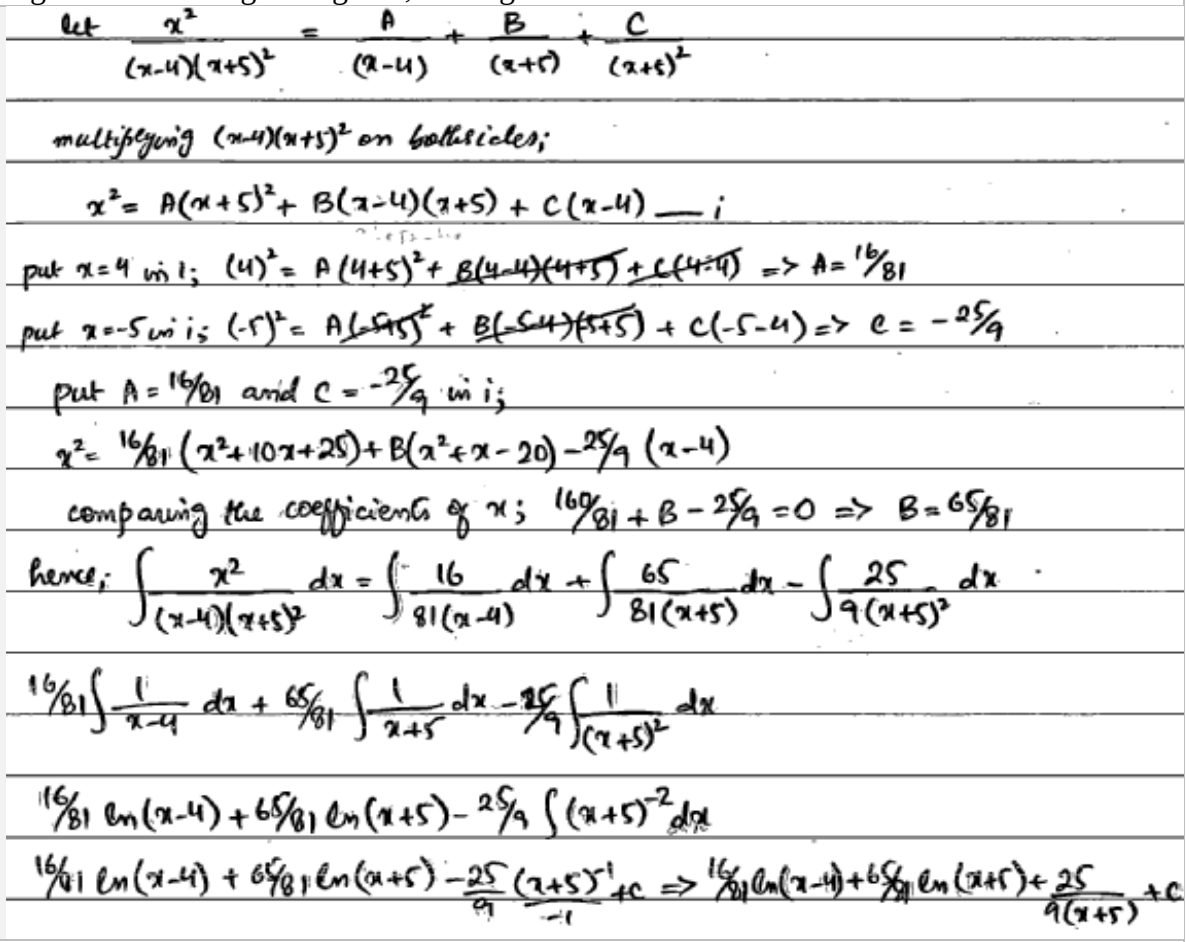
Any Additional Suggestion:

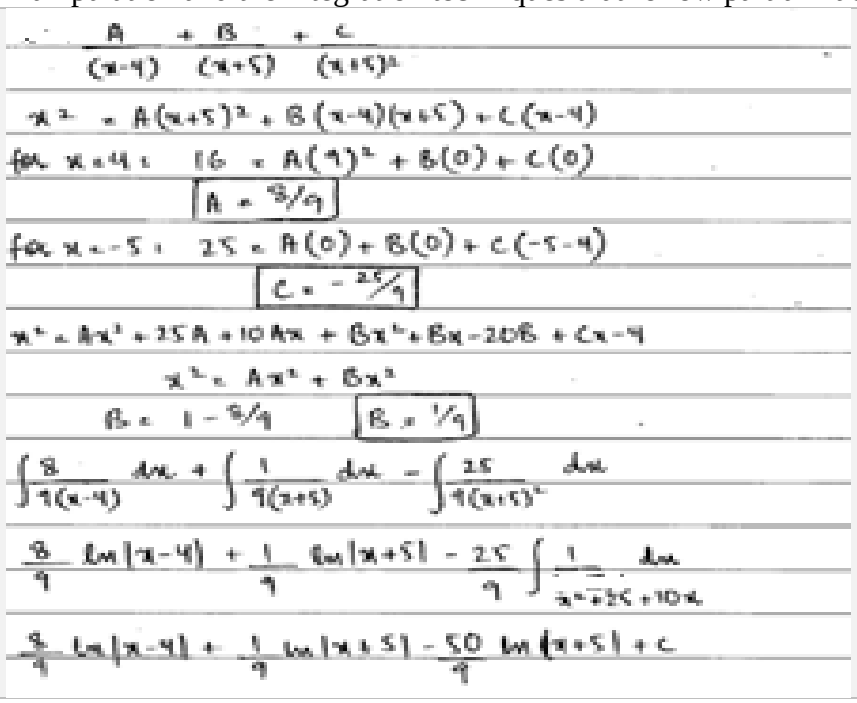
- Use a graphing tool to plot the original function and its integral to visualise the area under the curve.
- Give each group an integral that suggests a trig substitution (e.g., $\sqrt{(a^2-x^2)}$), and let them choose the correct substitution from a "substitution bank."
- Create a matching game with integrals on one side and their ideal trig substitutions on the other.
- Run a relay where one student performs the substitution, another integrates, and a third back-substitutes to the original variable.
- Provide real-world arc length or sector area problems that require trig substitution and have teams solve and compare answers.

Question No. 3b

Candidates were given the choice to attempt any TWO out of the three questions: 3a, 3b and 3c.

Question Text	Apply the method of partial fractions to evaluate the given integral. $\int \frac{x^2}{(x-4)(x+5)^2}\,dx$
SLO No.	16.5.1

SLO Text	Find $\int \frac{f(x)}{g(x)}dx$ using partial fractions where $f(x)$ and $g(x)$ are algebraic functions such that $g(x) \neq 0$.
Max Marks	5
Cognitive Level	A
Checking Hints	1 mark for writing the fraction as $\frac{A}{x-4}+\frac{B}{x+5}+\frac{C}{(x+5)^2}$ 1 mark for finding the values of A, B and C each (3 required) 1 mark for obtaining correct integral $\frac{16}{81}\ln(x-4)+\frac{16}{81}\ln(x-4)-\frac{25}{9(x+5)}+C$
Overall Performance	Overall, most candidates attempted this question. Many showed a good understanding of the method of partial fractions. However, some had trouble with the decomposition process, especially when dealing with repeated or more complex terms. Mistakes were also common during the solving and integration of the partial fractions.
Description of Better Responses	*Better responses* correctly recognised that the given rational function could be expressed as a sum of simpler fractions. These candidates accurately set up the partial fraction decomposition using the correct form with constants in the numerators. They carefully solved for the unknown values using either substitution or a system of equations. Once the decomposition was completed, they correctly integrated each term, applying the appropriate logarithmic or polynomial rules. Their solutions were well-organised and showed clear algebraic reasoning throughout, leading to the correct final result.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* often showed partial understanding. While some candidates were able to write the general form of the partial fractions, they struggled to solve for the constants correctly. Common issues included algebraic errors in solving the resulting equations or attempting shortcuts without proper working. In several cases, candidates completed the

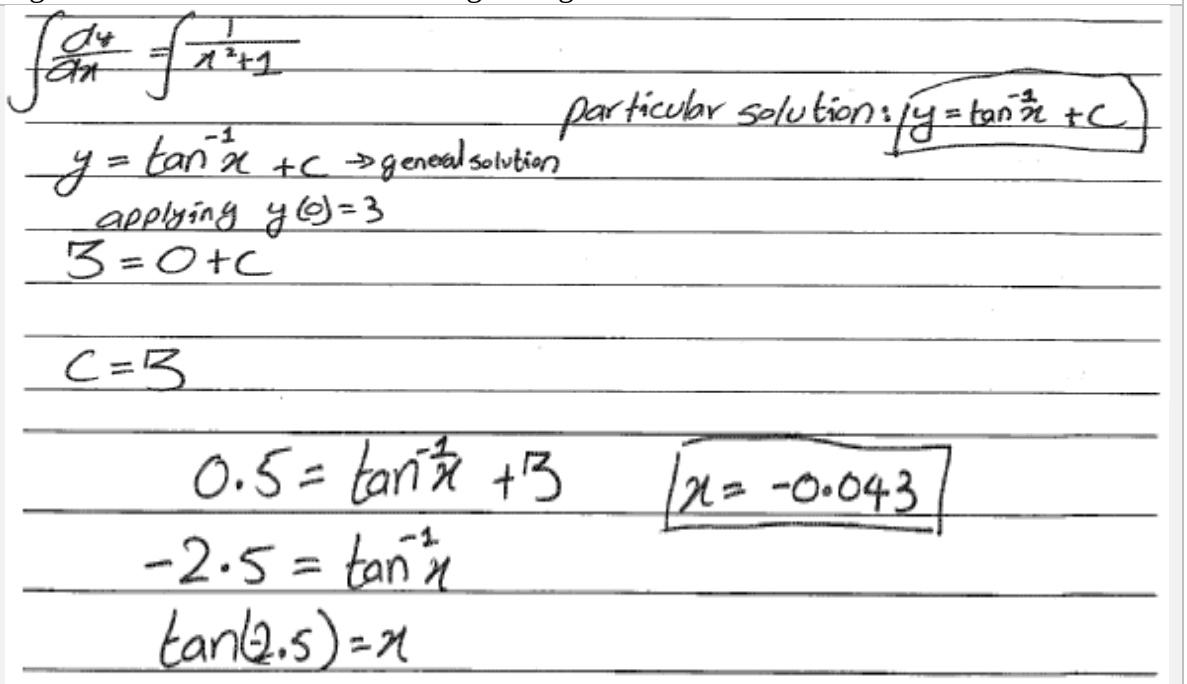
	decomposition accurately but failed to integrate the resulting terms correctly especially when dealing with logarithmic integrals. These errors reflected gaps in both algebraic manipulation and the integration techniques that follow partial fraction expansion.
Image of Weaker Response	

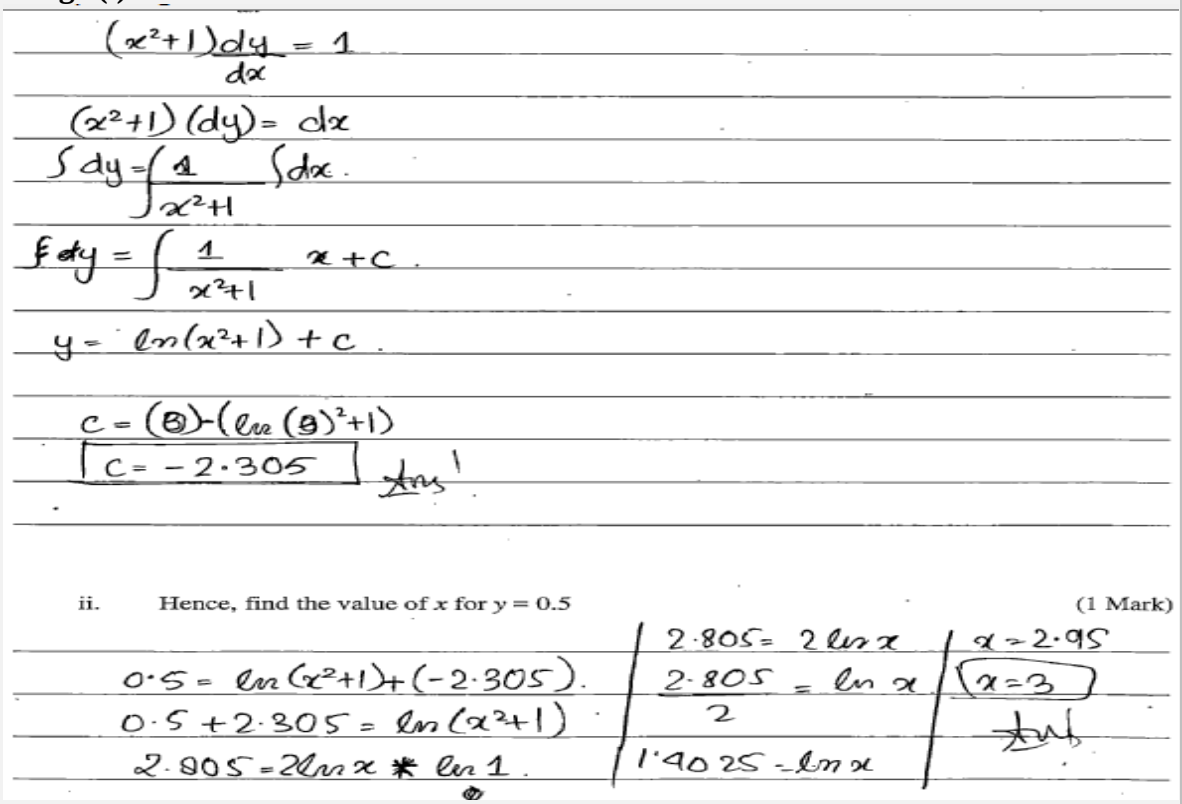
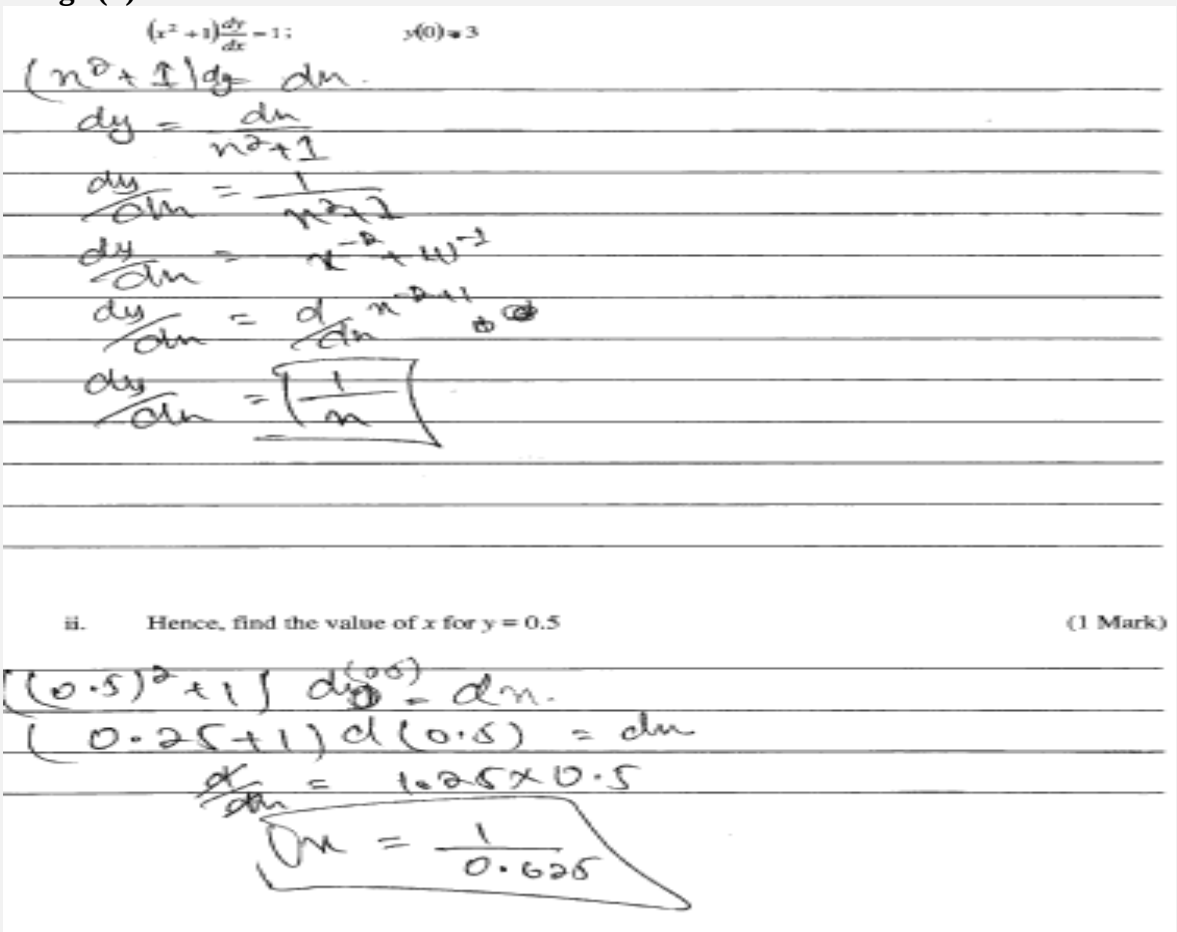
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Give students a table of *u* and *dv* choices for various integrals; they select the most efficient pair and justify their choice.
- Use a matching game where one card shows an integral and the other shows the correct *u* – *dv* breakdown.
- Run a relay where one student finds *u* and *dv*, the next computes *du* and *v*, and the last assembles the integration by parts formula.
- Provide real-world scenarios (e.g., work done by variable force) and have students model and solve them using partial integration.
- Challenge students with "integration by parts twice" problems and let them work in pairs to avoid common mistakes.

Question No. 3c Candidates were given the choice to attempt any TWO out of the three questions: 3a, 3b and 3c.	
Question Text	i. Find the particular solution of the given differential equation. $\left(x^2+1\right)\frac{dy}{dx}=1;\qquad y(0)=3$ ii. Hence, find the value of x for $y = 0.5$
SLO No.	16.7.2
SLO Text	Solve differential equations of first order and first degree by separating the variables.
Max Marks	5
Cognitive Level	A
Checking Hints	i. 1 mark for separating variables and writing in the form $dy=\frac{1}{\left(x^2+1\right)}dx$ 1 mark for integrating to obtain $y=\tan^{-1}x+C$ as general solution 1 mark for substituting the initial conditions and evaluating C 1 mark for finding the particular solution by substituting the value of C ii. 1 mark for finding the value of x for the given value of y
Overall Performance	Overall, student performance on this question was satisfactory. Many candidates correctly identified the equation as a standard first-order differential equation and used basic integration methods to solve it. Most showed they understood how to separate variables and apply the initial condition to find the particular solution. However, several candidates struggled with the integration step or forgot to include the constant of integration, which led to incomplete or incorrect answers.
Description of Better Responses	*Better responses* demonstrated a strong understanding of how to approach the differential equation. Candidates correctly rearranged the equation to isolate variables and applied integration to both sides. They included the constant of integration and used the given condition $y(0)$ to find its value. Their solutions were laid out, showing each step with proper reasoning and mathematical accuracy. In the second part, these candidates successfully substituted the value y = 0.5 into the particular solution and solved for x, showing strong algebraic skills and accurate working throughout.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed several common difficulties. Some candidates struggled to rearrange the equation into a form suitable for integration, while others incorrectly integrated one or both sides. A frequent issue was the omission of the constant of integration

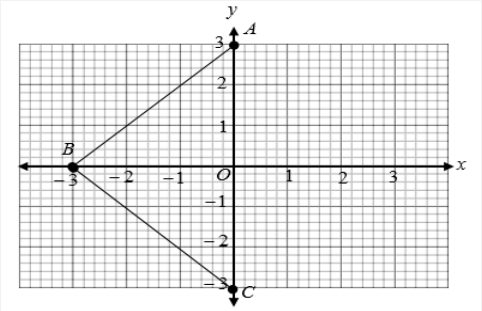
	or failure to use the initial condition to determine it. In a few cases, the integration was done correctly, but candidates did not go on to solve for x in part (ii), leaving the response incomplete. These issues point to a need for further practice in solving basic first-order differential equations and applying initial conditions properly.
Images of Weaker Responses	**Image (i)**  $(x^2+1)\frac{dy}{dx} = 1$ $(x^2+1)(dy) = dx$ $\int dy = \int \frac{1}{x^2+1} \int dx.$ $\int dy = \int \frac{1}{x^2+1} x + c.$ $y = \ln(x^2+1) + c.$ $c = (0) - (\ln(0)^2+1)$ $c = -2.305$ ~~1m~~ 1 ii. Hence, find the value of x for $y = 0.5$ (1 Mark) $0.5 = \ln(x^2+1) + (-2.305)$ $0.5 + 2.305 = \ln(x^2+1)$ $2.805 = 2\ln x * \ln 1$ $2.805 = 2\ln x$ $\frac{2.805}{2} = \ln x$ $1.4025 = \ln x$ $x = 2.95$ $x = 3$ ~~1m~~
	Image (ii)  $(x^2+1)\frac{dy}{dx} = 1; \quad y(0) = 3$ $(x^2+1)dy = dx.$ $dy = \frac{dx}{x^2+1}$ $\frac{dy}{dx} = \frac{1}{x^2+1}$ $\frac{dy}{dx} = x^{-2} + 1)^{-1}$ $\frac{dy}{dx} = \frac{d}{dx} x^{-2+1}$ $\frac{dy}{dx} = \frac{1}{x}$ ii. Hence, find the value of x for $y = 0.5$ (1 Mark) $(0.5)^2+1)\, dy^{(0.5)} = dx.$ $(0.25+1)\, d(0.5) = dx$ $\frac{d}{dx} = 1.25 \times 0.5$ $x = \frac{1}{0.625}$

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Give students real-life growth or decay problems (e.g., population, cooling) and have them model and solve by separating variables.
- Use a matching game where one card shows a differential equation and the other shows its separated form.
- In pairs, students race to rearrange mixed-variable equations into separated form before integrating.
- Use graphing tools to plot both the slope field and the solution curve after solving the equation.

Question No. 4	
Question Text	In the given diagram, *ABC* is a triangle with the given vertices on the Cartesian coordinate system. Using the given information, find the i. equation of the side *AB*. ii. equation of the altitude from C to *AB*. 
SLO No.	17.7.2
SLO Text	Find the equation of median, altitude and right bisector of a triangle.
Max Marks	4
Cognitive Level	
Checking Hints	i. 1 mark for substituting the coordinates in the determinant form of the equation/ 1 mark for correct calculation of slope of AB/ 1 mark for correct calculation of intercepts a/b 1 mark writing the equation ii. 1 mark for finding the slope of the altitude as -1 1 mark for finding the equation of the altitude
Overall Performance	Overall, candidates struggled with this question, although some demonstrated a good understanding of straight-line equations, particularly in part (i). A number of candidates were able to correctly use the given coordinates to form the required equations, while others had difficulty accurately identifying the coordinates of the vertices or applying the correct formula to find the equation of a line. Part (ii), which involved understanding perpendicular slopes, proved more challenging for most candidates.

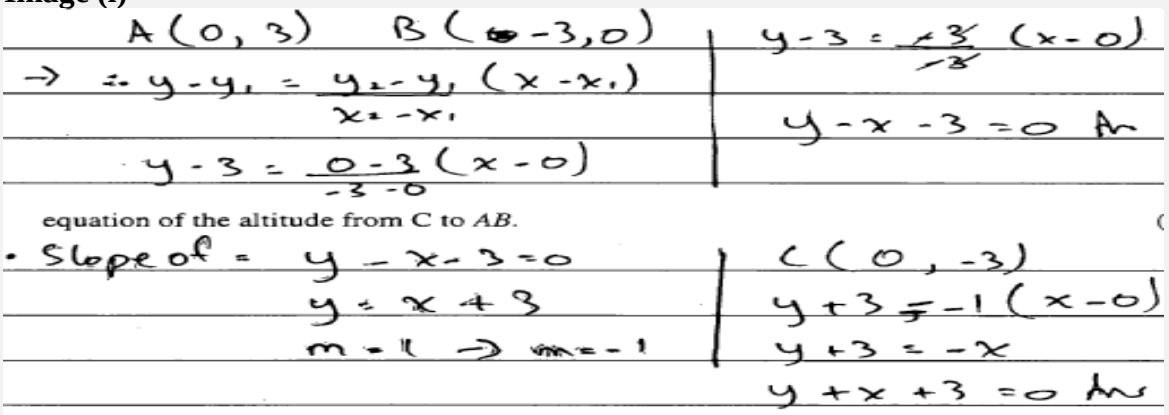
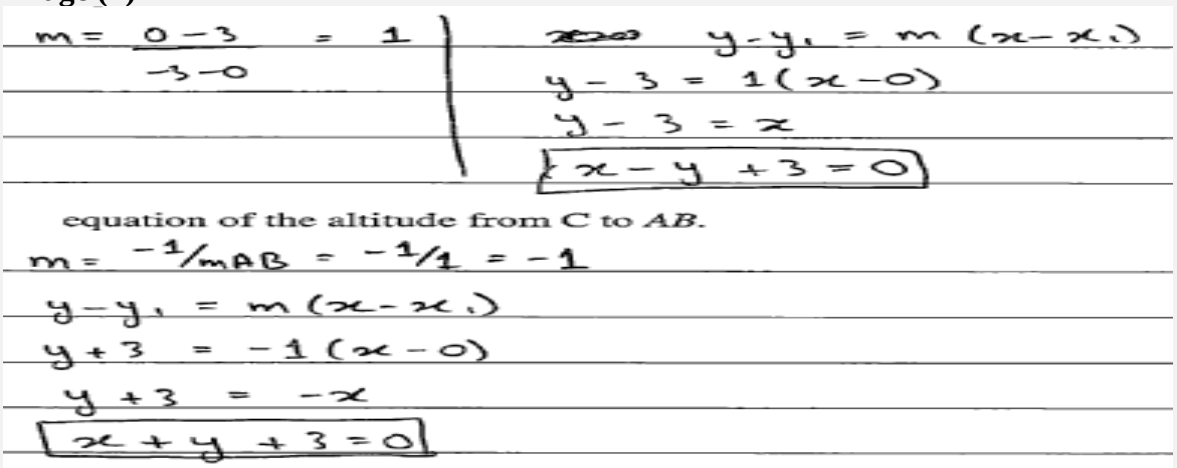
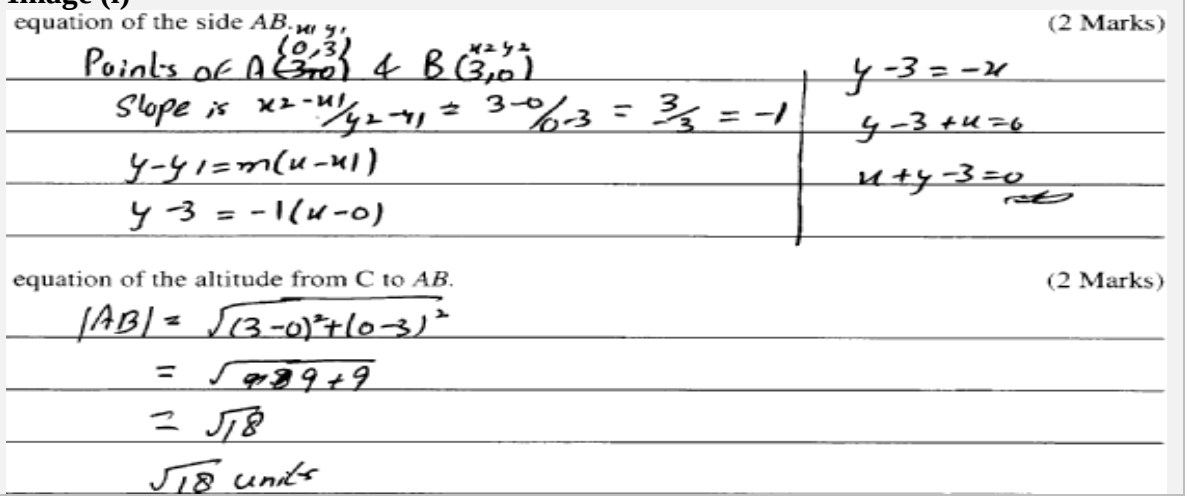
Description of Better Responses	*Better responses* showed clear identification of the given points A, B, and C. These candidates used either the two-point form or slope-intercept form accurately to find the equation of line AB. In part (ii), they correctly calculated the gradient of AB and then found the negative reciprocal to determine the slope of the altitude from point C. They used this slope with point C to form the equation of the perpendicular line. Their responses were well-organised and mathematically sound, showing a clear grasp of coordinate geometry concepts.
Images of Better Responses	**Image (i)**  A(0, 3) B(-3, 0) → ∴ $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$ $y - 3 = \frac{0 - 3}{-3 - 0}(x - 0)$ $y - 3 = \frac{-3}{-3}(x - 0)$ $y - x - 3 = 0$ Ans equation of the altitude from C to *AB*. • Slope of $y - x - 3 = 0$ $y = x + 3$ $m = 1 \rightarrow m = -1$ C(0, -3) $y + 3 = -1(x - 0)$ $y + 3 = -x$ $y + x + 3 = 0$ Ans **Image (ii)**  $m = \frac{0 - 3}{-3 - 0} = 1$ $y - y_1 = m(x - x_1)$ $y - 3 = 1(x - 0)$ $y - 3 = x$ $x - y + 3 = 0$ equation of the altitude from C to *AB*. $m = -1/m_{AB} = -1/1 = -1$ $y - y_1 = m(x - x_1)$ $y + 3 = -1(x - 0)$ $y + 3 = -x$ $x + y + 3 = 0$
Description of Weaker Responses	*Weaker responses* often showed confusion in identifying or using the correct coordinates. Some candidates interchanged points or misread them from the diagram. In applying straight-line equations, several responses incorrectly used the slope formula or substituted values inaccurately. For the altitude, many failed to recognise the need for the perpendicular slope and instead used the same slope as *AB* or did not apply the correct method to find the equation from point *C*. These errors led to incomplete or incorrect answers.
Images of Weaker Responses	**Image (i)**  equation of the side *AB*. (2 Marks) Points of A(0,3) & B(3,0) Slope is $x_2 - x_1 / y_2 - y_1 = 3 - 0 / 0 - 3 = 3/-3 = -1$ $y - y_1 = m(x - x_1)$ $y - 3 = -1(x - 0)$ $y - 3 = -x$ $y - 3 + x = 0$ $x + y - 3 = 0$ equation of the altitude from C to *AB*. (2 Marks) $\|AB\| = \sqrt{(3-0)^2 + (0-3)^2}$ $= \sqrt{9 + 9}$ $= \sqrt{18}$ $\sqrt{18}$ units

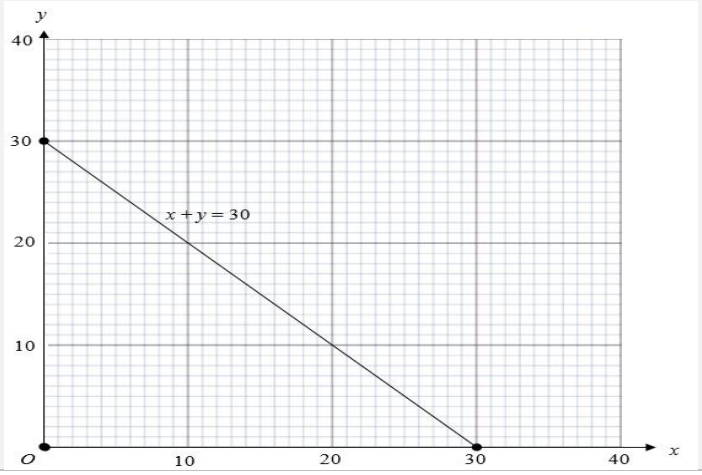
	Image (ii) equation of the side AB. $A=(0,3)$ $B=(-3,0)$ $\|AB\| = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ $= \sqrt{(-3-0)^2+(0-3)^2}$ $= \sqrt{9+9}$ $= \sqrt{18}$

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- Use GeoGebra to plot a triangle from given coordinates and visually draw its median, altitude, and perpendicular bisector, then verify equations algebraically.
- Create a "geometry to algebra" matching game where one card shows a triangle with a highlighted line, and another shows its equation.
- Give each group different triangle coordinates and ask them to find all three special lines, then compare results with graphing tool outputs.
- Run a relay where one student finds the midpoint, another finds slope, and the third writes the final equation for the given line type.
- Provide real-life application tasks (e.g., finding balance point of a triangular plate) for students to solve using median equations.

Question No. 5		
Question Text	Using the given graph, find the maximum value of the objective function by shading the feasible region in the given linear programing problem. $P(x, y) = 300x + 200y$ $x + y \leq 30$ $x + 2y \leq 40$ $x \geq 0,\ y \geq 0$	

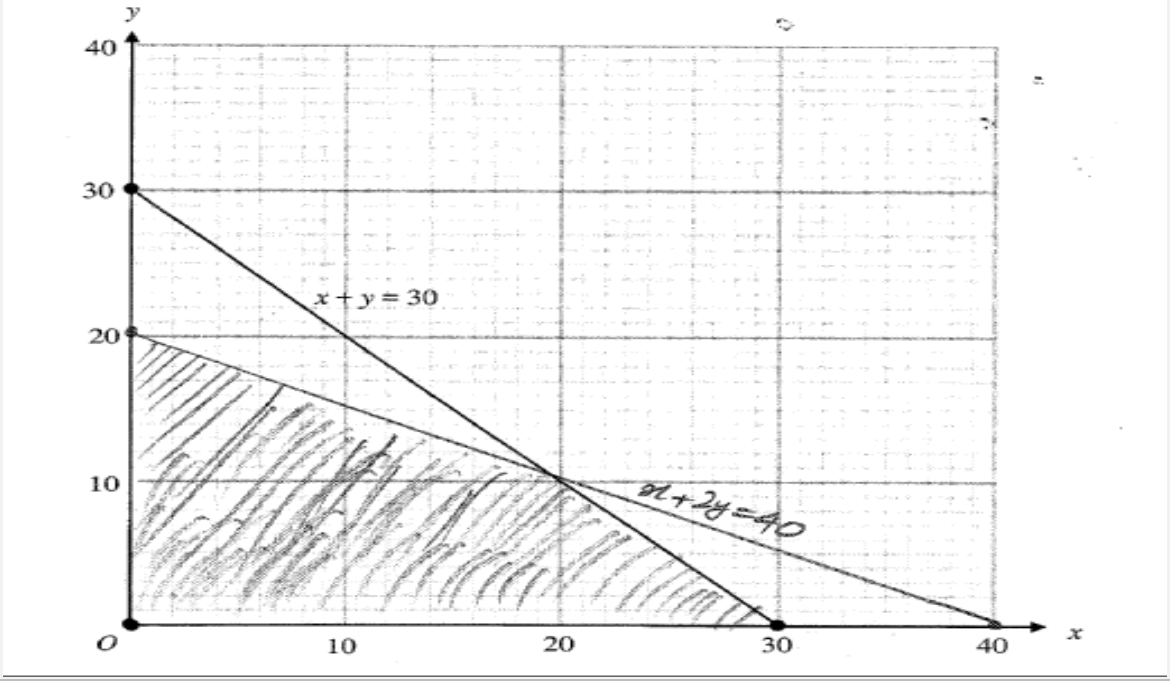
SLO No.	18.4.2
SLO Text	Find the optimal solution through the following systematic procedure: a. establish the mathematical formulation of linear programing problems, b. construct a graph, c. identify the feasible region, d. locate the solution points, e. evaluate the objective function, f. select the optimal solution, g. verify the optimal solution by actually substituting values of variables from the feasible region.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for drawing the line representing the inequality 1 mark for identifying/shading the solution regions 1 mark for identifying the corner points 1 mark for evaluating the maximum value of the objective function
Overall Performance	Overall, most candidates solved this question partially correctly. Some were able to draw the first inequality and identify parts of the feasible region. However, only a few of the candidates completed all the steps accurately and reached the correct maximum value of the objective function by identifying the correct corner points.
Description of Better Responses	*Better responses* showed a sound understanding of the terminology and method used in linear programming. These candidates correctly plotted the inequalities on the graph, shaded the feasible region, and identified the corner points of the region. They accurately substituted each corner point into the objective function and selected the value that gave the maximum result. Their responses demonstrated a clear, step-by-step approach and a firm understanding of how to link algebraic constraints to graphical interpretation and optimisation.
Images of Better Responses	**Image (i)** corner points: (0,20), (30,0), (20,10) $x+y=30$, $x+2y=40$ $x=30-y$: $30-y+2y=40$ $x=20$ $30+y=40$ $y=10$ $P(x,y) = 300(20)+200(10)$ $= 6000+2000$ $= 8000$ → max value 

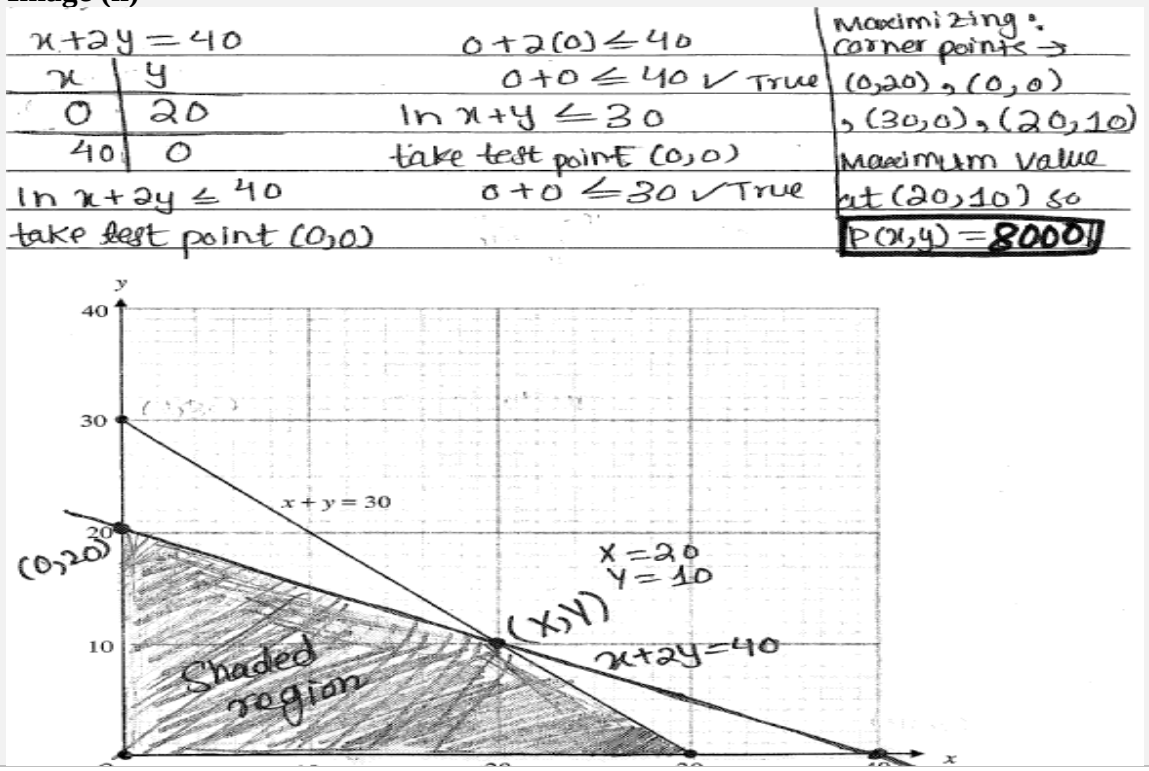
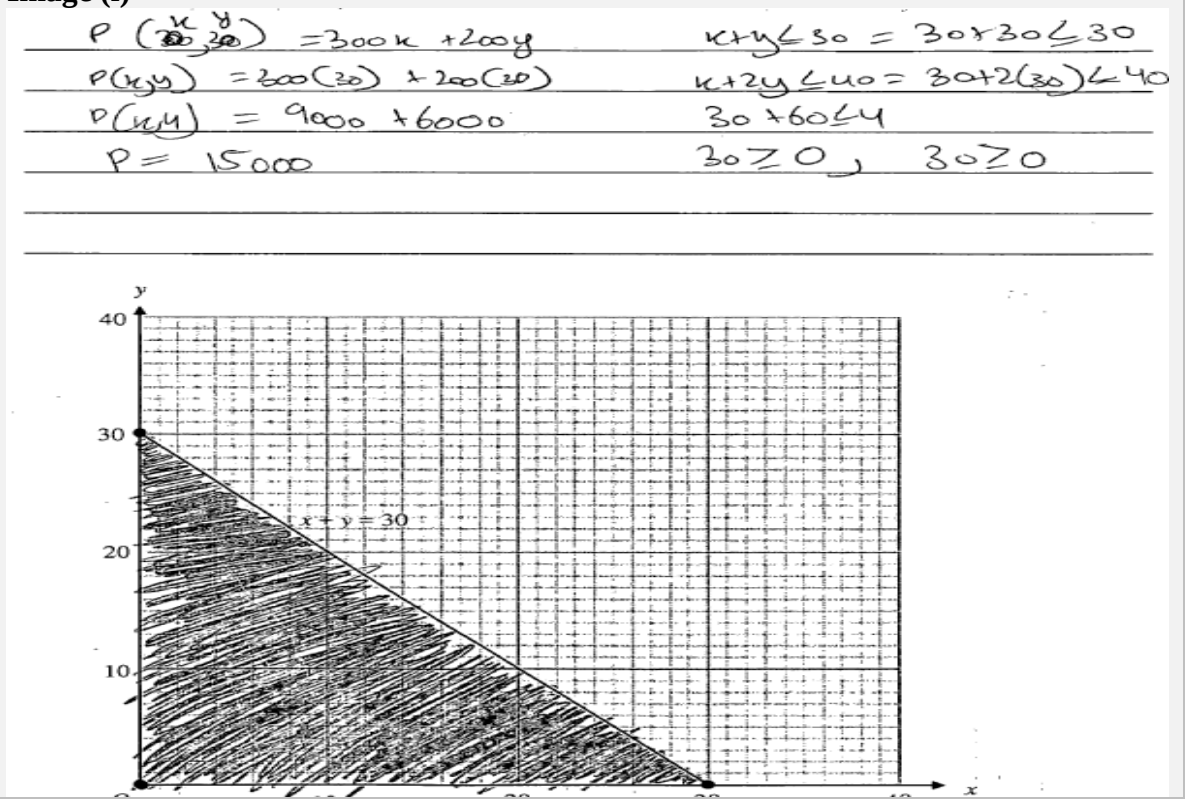
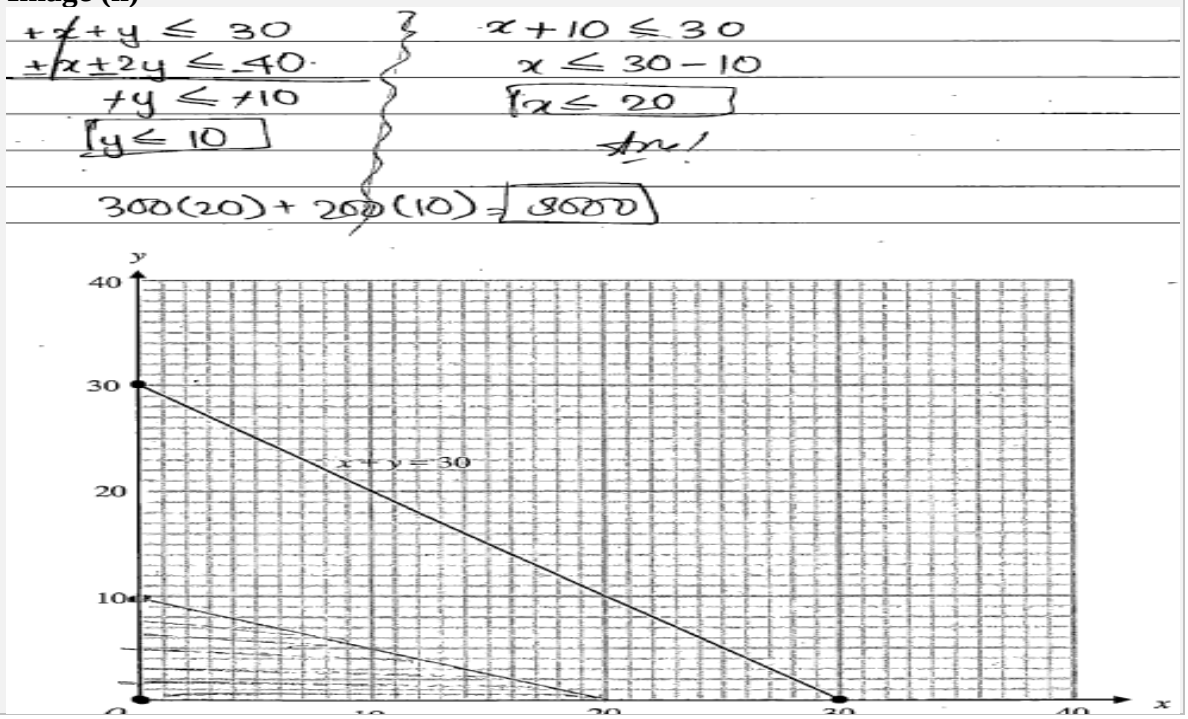
	Image (ii) 
Description of Weaker Responses	*Weaker responses* often showed confusion with basic linear programming steps. Several candidates failed to identify or correctly draw the lines for all inequalities, most commonly omitting or misrepresenting the second constraint $x+2y \leq 40$. Others shaded incorrect regions or failed to define the feasible region clearly. A common error was to ignore or incorrectly identify corner points, which are essential to evaluating the objective function. Many candidates also did not substitute these points into the objective function or did so incorrectly. This indicates a lack of understanding of both the graphical and algebraic aspects of the topic.
Images of Weaker Responses	**Image (i)** 

Image (ii)

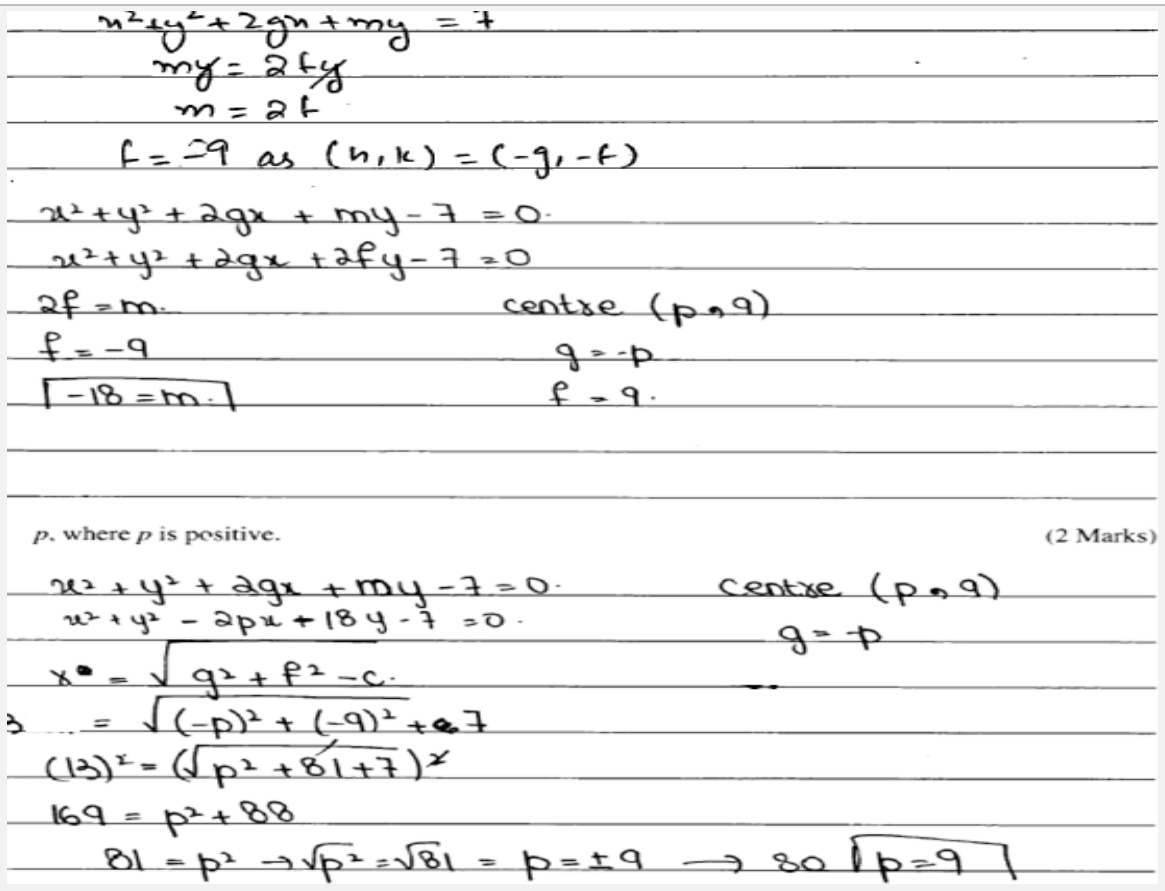


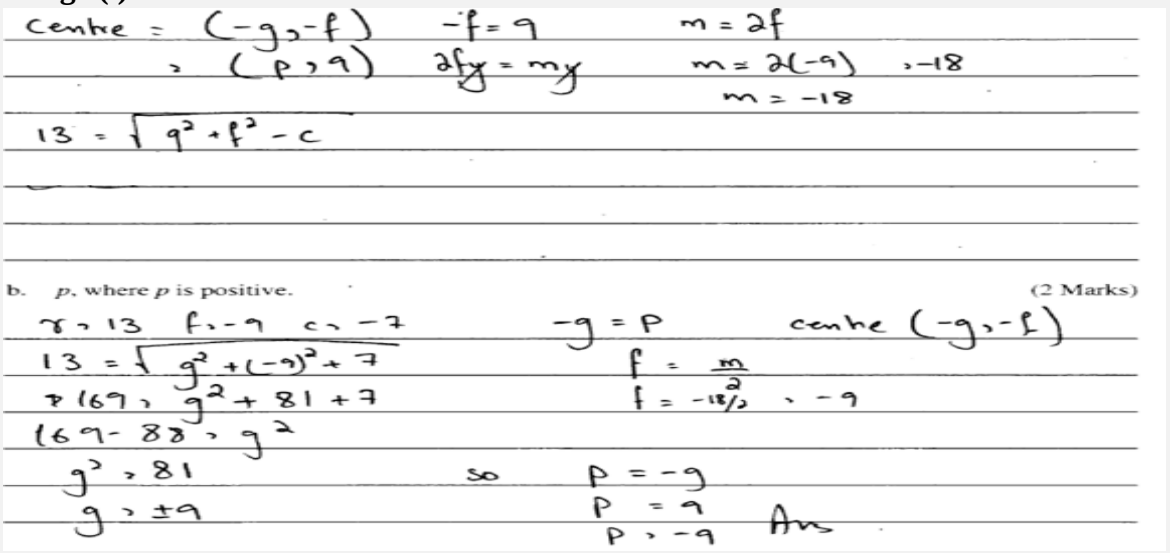
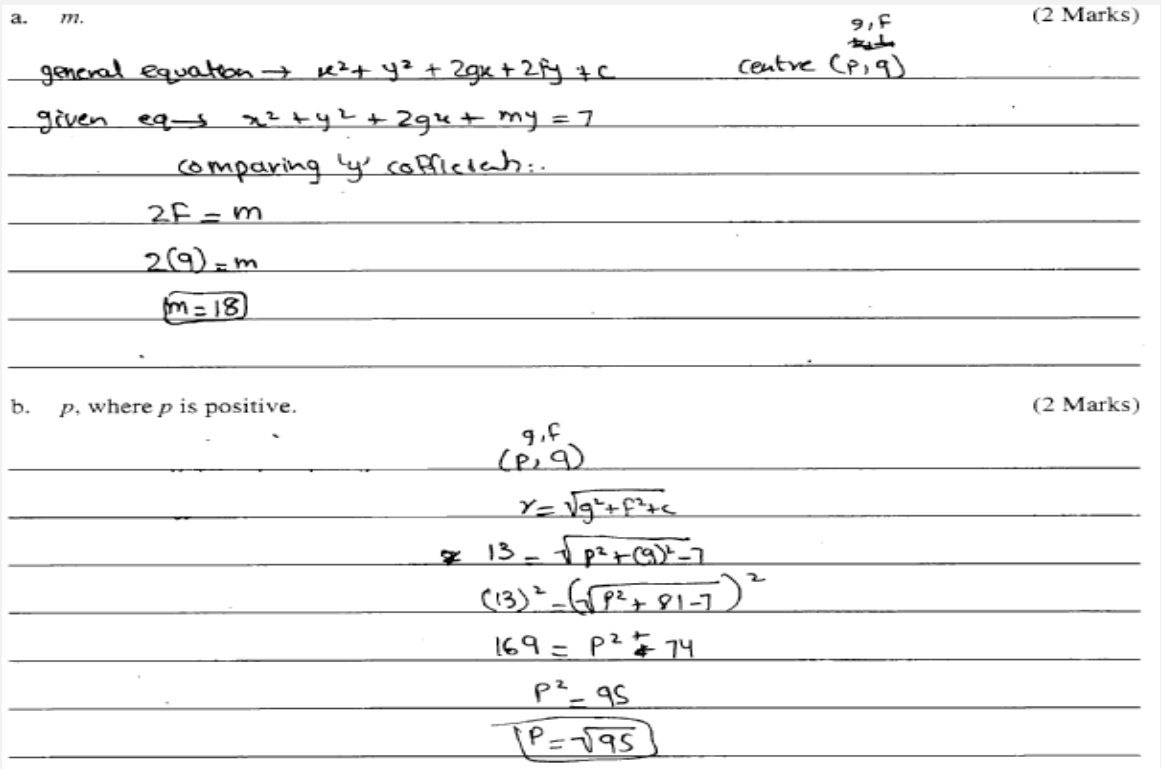
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Give real-life scenarios (profit maximisation, cost minimisation) and have students write the corresponding mathematical formulation.
- Use a graphing tool or graph paper for students to plot constraints and visually construct the feasible region.
- Organise a "feasible or not?" challenge where students quickly decide if given points lie inside the feasible region.
- In pairs, students locate all corner points of the feasible region and label them clearly on the graph.
- Students evaluate the objective function at each corner point and record the results in a table.
- Run a "fastest optimiser" contest where teams select the optimal point and justify why it is the best solution.
- Have students verify their chosen optimal solution by substituting values into all original constraints to check feasibility.

Question No. 6	
Question Text	The general equation of a circle is given as $x^2 + y^2 + 2gx + my = 7$. If the centre of the circle is (p, 9) and radius is 13 units, then find the value of a. m. b. p, where p is positive.
SLO No.	19.2.4
SLO Text	Find the centre and radius by using the general and standard form of equation of a circle.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for writing $f = -9$ 1 mark for finding m ii. 1 mark for substituting the values in $r = \sqrt{g^2 + f^2 - c}$ 1 mark for finding the value of p
Overall Performance	Overall, candidates generally found this question challenging. A few were able to correctly use the properties of the standard form of a circle's equation, but most either answered only part of the question or failed to make the right connections. The responses indicated a weak understanding of how to identify and use key parameters like the centre and radius from the general equation.
Description of Better Responses	*Better responses* showed a clear understanding of the structure of the general equation of a circle and how to relate it to its geometric properties. Candidates correctly identified that the centre of a circle in the general form. From this, they compared the given values and correctly found that $f = -9$, leading to the accurate determination of $m = -18$. They then used the centre (p , 9) and the given radius to substitute into the standard form of the circle equation and solve for the value of p, selecting the correct positive value. Their work was methodical and well-reasoned.
Image of Better Response	

Description of Weaker Responses	*Weaker responses* revealed several key misconceptions. Many candidates failed to correctly match the given general equation to the standard form of a circle. Some incorrectly identified or compared coefficients, leading to errors in determining the values of *g*, *f*, and mmm. Others substituted the given point into the equation but did so inaccurately or without recognising the role of the radius in solving for the unknown *p*. A number of candidates left the response incomplete or provided unclear working, indicating uncertainty in applying the concepts.
Images of Weaker Responses	**Image (i)**  Centre = $(-g,-f)$ $-f=9$ $m=2f$ $=(p,9)$ $2fy=my$ $m=2(-9)$ -18 $m=-18$ $13=\sqrt{g^2+f^2-c}$ b. *p*, where *p* is positive. (2 Marks) $r=13$ $f=-9$ $c=-7$ $-g=p$ centre $(-g,-f)$ $13=\sqrt{g^2+(-9)^2+7}$ $f=\frac{m}{2}$ $169=g^2+81+7$ $f=-18/2=-9$ $169-88=g^2$ $g^2=81$ so $p=-g$ $g=\pm 9$ $p=9$ Ans $p=-9$ **Image (ii)**  a. *m*. (2 Marks) general equation → $x^2+y^2+2gx+2fy+c$ centre $(p,9)$ g,f given eq → $x^2+y^2+2gx+my=7$ comparing 'y' coefficients: $2f=m$ $2(9)=m$ $m=18$ b. *p*, where *p* is positive. (2 Marks) $(p,9)$ g,f $r=\sqrt{g^2+f^2+c}$ $13=\sqrt{p^2+(9)^2-7}$ $(13)^2=(\sqrt{p^2+81-7})^2$ $169=p^2+74$ $p^2=95$ $p=\sqrt{95}$

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone	• Past paper questions • Discussion on E-Marking Notes

• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Use GeoGebra to plot circles from both general and standard forms, then visually identify the centre and radius before calculating them.
- Create a matching game where one card shows an equation and another shows its correct centre–radius pair.
- Give students incomplete square steps in the conversion from general to standard form and have them fill in the blanks.
- Organise a relay where one student completes the square, another identifies the centre, and the third calculates the radius.
- Provide mixed equations and have students quickly sort them into "already standard form" and "needs conversion" piles.

Question No. 7a

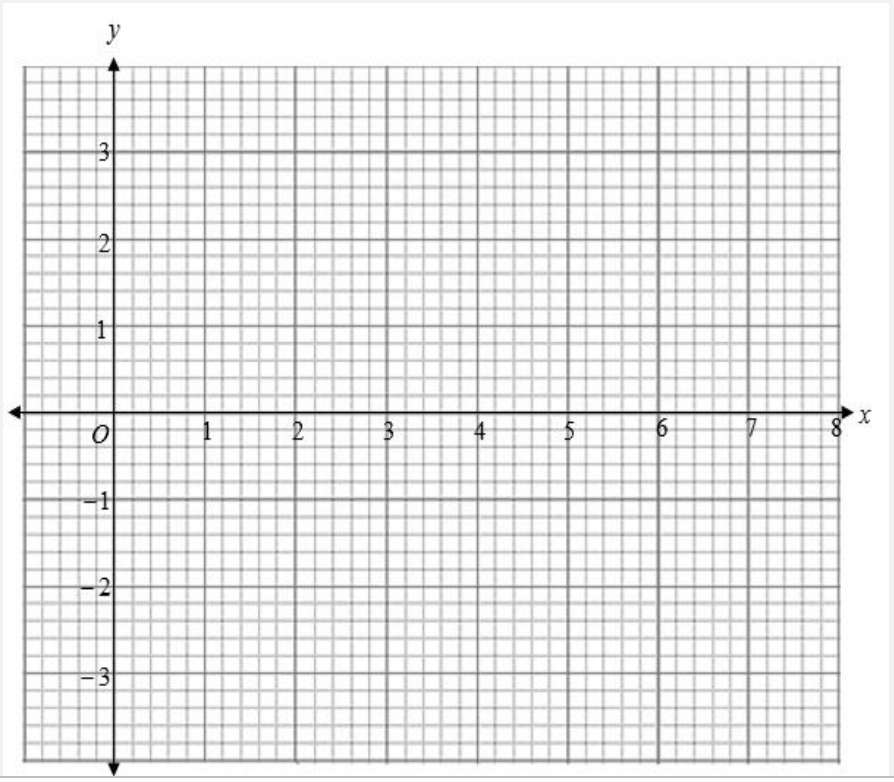
Candidates were given the choice to attempt any TWO out of the three questions: 7a, 7b and 7c.

Question Text

A parabola is represented by the equation $y^2 = 12(x-2)$.

i. Find the following elements of the parabola.
   I. Vertex
   II. Focus

ii. Sketch the parabola and label the elements found in part i.



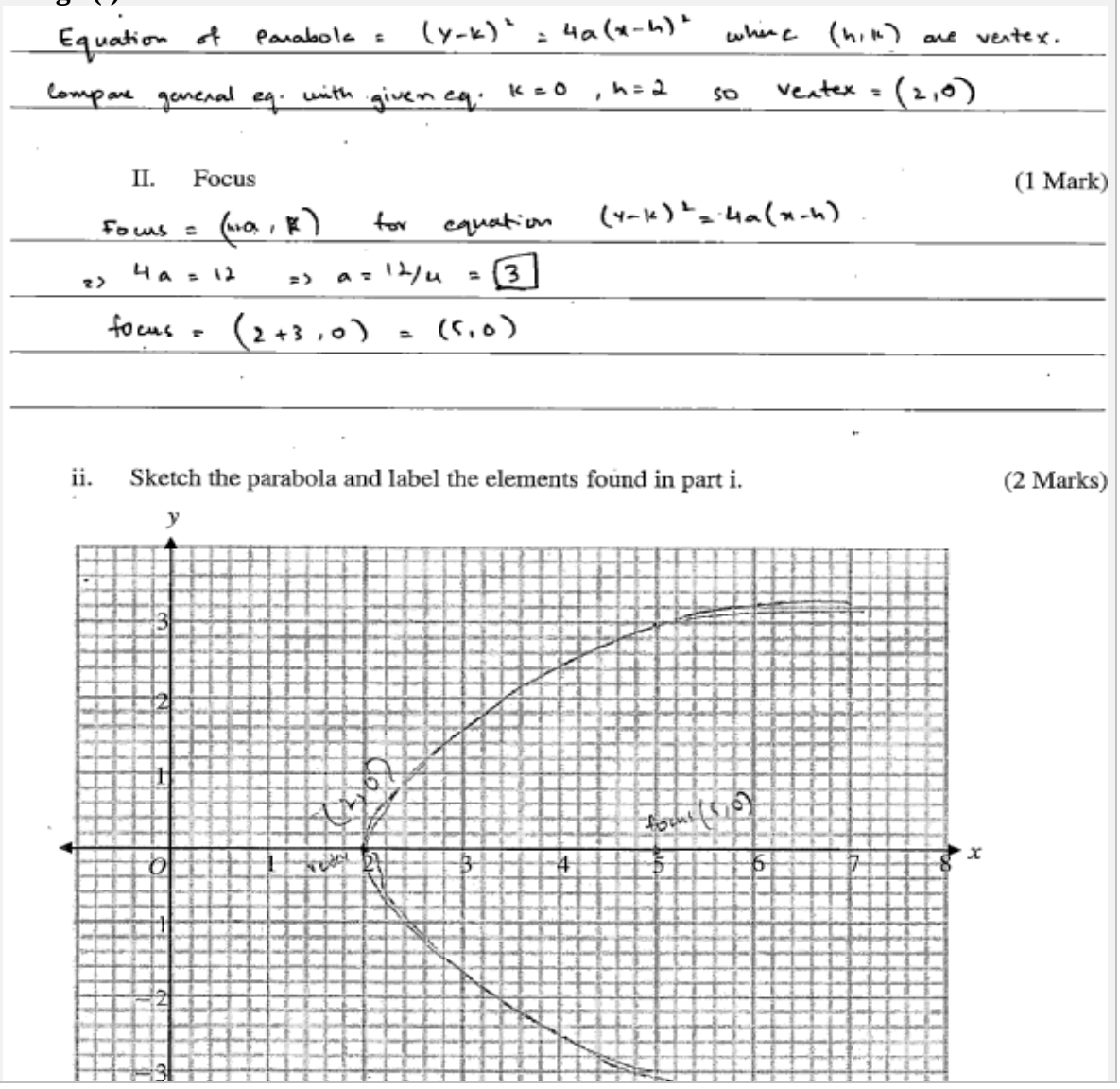
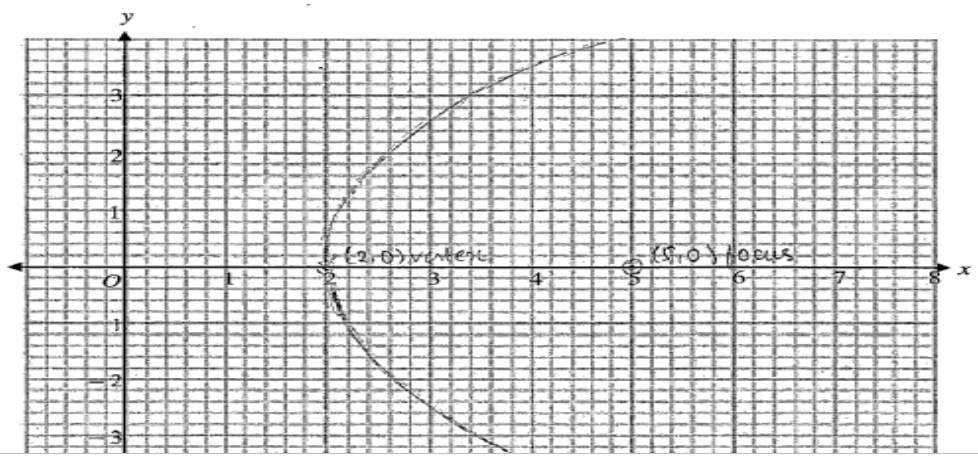
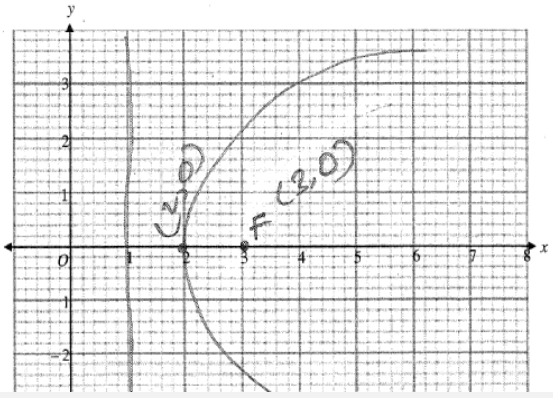
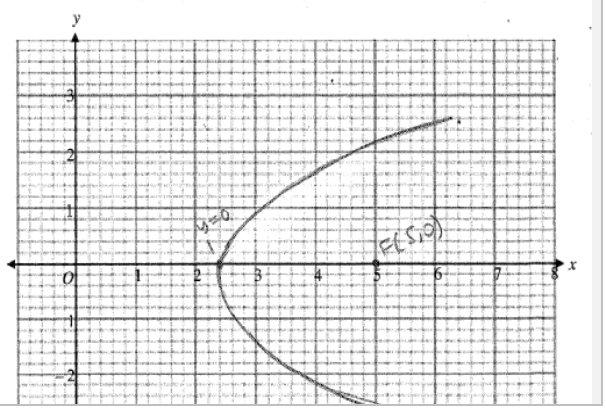
SLO No.	20.2.2 and 20.2.3
SLO Text	Find the elements of a parabola. Sketch a parabola with given elements.
Max Marks	4
Cognitive Level	U
Checking Hints	i. 1 mark for finding focus 1 mark for finding vertex ii. 1 mark for correct sketching (opens right) 1 mark for rightly showing vertex
Overall Performance	Overall, most candidates attempted this question, but only a few managed to score full marks. Some showed a good understanding of the structure of a parabola and were able to correctly find either the vertex or the focus. However, many responses were only partly correct. A number of candidates struggled to apply the standard forms and calculate the required elements accurately.
Description of Better Responses	Candidates who performed well recognised the given equation as that of a standard horizontal parabola. They correctly identified the vertex as (2,0) by comparing the equation with the standard form $(y - k)^2 = 4a(x - h)$. They accurately found the value of $a = 3$ and used it to determine the coordinates of the focus as $(2 + 3, 0) = (5, 0)$. Their sketches were labelled clearly, showing the vertex, focus, and the correct orientation of the parabola opening to the right.
Images of Better Responses	**Image (i)**  Equation of parabola : $(y-k)^2 = 4a(x-h)^2$ where (h,k) are vertex. Compare general eq. with given eq. $k=0$, $h=2$ so vertex $= (2,0)$ II. Focus (1 Mark) Focus $= (4a, k)$ for equation $(y-k)^2 = 4a(x-h)$ $\Rightarrow 4a = 12 \Rightarrow a = 12/4 = \boxed{3}$ focus $= (2+3, 0) = (5, 0)$ ii. Sketch the parabola and label the elements found in part i. (2 Marks) y x O vertex (2,0) focus (5,0)

	Image (ii) $\therefore (y-k)^2 = 4a(x-h)$ and vertex $= (h,k)$ so vertex $= (2,0)$ II. Focus (1 Mark) focus $= (h+a, k)$ and $4a = 12 \Rightarrow a = 3$ hence focus $= (2+3, 0)$ $= (5,0)$ ii. Sketch the parabola and label the elements found in part i. (2 Marks) 
Description of Weaker Responses	*Weaker responses* revealed limited understanding of how to extract elements from the standard equation of a parabola. Many candidates confused the general form with vertical parabolas or incorrectly identified the vertex as the origin. Several did not understand how to calculate the focus using the value of *a*, and some omitted the sketch or drew a parabola in the wrong direction. Errors were also seen in placing and labelling key points, indicating gaps in concept application and interpretation.
Images of Weaker Responses	**Image (i)** I. Vertex (2,0) II. Focus $4a = 12$ $a = 3$ (3,0) ii. Sketch the parabola and label the elements found in part i.  **Image (ii)** I. Vertex Vertex of parabola $y^2 = 12(x-2)$ is $y=0$ II. Focus $(\pm c, 0) \Rightarrow y^2 = 12(x-2) \Rightarrow 4a = 12$ $a = 3$, $h = +2$ $\Rightarrow$ focus $= (h+a, k) \Rightarrow (2+3, 0) \Rightarrow (5,0)$ $(5,0) \rightarrow$ focus of parabola ii. Sketch the parabola and label the elements found in part i. 

Suggestions for improvement (Highlight all that apply)

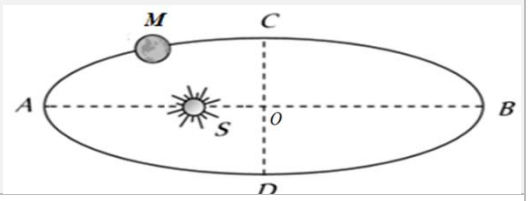
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

- Use GeoGebra for students to plot a parabola given its vertex, focus, and directrix, then label all elements on the graph.
- Give a "parabola parts" matching set where cards with definitions (vertex, focus, axis, directrix, latus rectum) are matched to diagrams.
- Provide equations of parabolas and have students identify all elements before sketching.
- Run a "sketch race" where students are given elements (vertex, focus, directrix) and must draw the correct parabola on graph paper.
- Challenge students to change one element (e.g., move the focus) and predict how the parabola's shape and position will change before verifying on a graphing tool.

Question No. 7b

Candidates were given the choice to attempt any TWO out of the three questions: 7a, 7b and 7c.

Question Text	The planet Mars revolves around the Sun in an elliptic orbit. During its revolution, Mars keeps the minimum distance of 129.5 million kilometres and maximum distance of 154.4 million kilometres from the Sun. The situation is depicted in the given diagram. Assuming the Sun as on one of the foci, find the i. location of the Sun. ii. equation representing the orbit of Mars. 
SLO No.	21.2.5
SLO Text	Solve problems related to ellipse.
Max Marks	4
Cognitive Level	A
Checking Hints	1 Mark for each correctly identifying the maximum and minimum distance on ellipse 1 Mark for finding the value of c 1 Mark for finding the value of b 1 Mark for substituting in the standard equation of ellipse
Overall Performance	Overall, only a small number of candidates attempted this question. Among those who did, several gave partially correct responses. The question relied on understanding the properties and equation of an ellipse, which many candidates found difficult to recall and apply correctly.

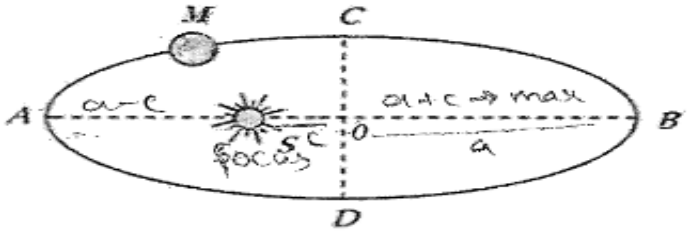
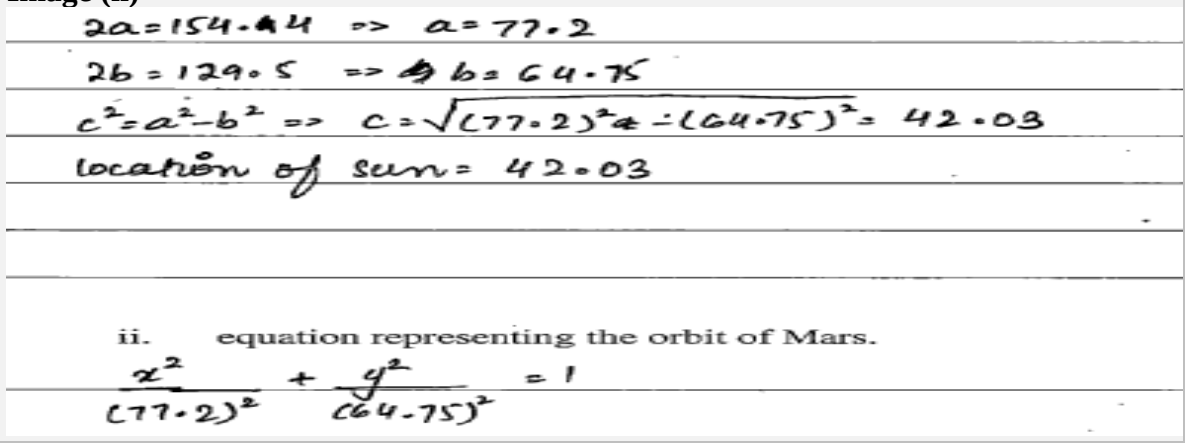
Description of Better Responses	*Better responses* showed a clear understanding of the terminology and key elements of an ellipse. Candidates correctly identified that the minimum and maximum distances represent the perihelion and aphelion, respectively, used to determine the values of the semi-major axis *a* and the distance from the centre to the focus *c*. They used the relationship $c^2 = a^2 - b^2$ and correctly formed the standard equation of a horizontal ellipse. The location of the Sun was identified as one of the foci, and their final equation accurately reflected the geometry of the orbit. These candidates demonstrated logical reasoning and accurate calculations.
Image of Better Response	 Assuming the Sun as on one of the foci, find the i. location of the Sun. (2 Marks) max distance a+c = 154.4 min distance a-c = 129.5 2a = 141.95 c = 154.4 - 141.95 c = 12.45 km → location of Sun from center. million kilometers 129.5 million km → Sun's distance from nearest. ii. equation representing the orbit of Mars. (2 Marks) $\sqrt{a^2-c^2} = b$ 141.4 = b $\frac{x^2}{(141.95)^2} + \frac{y^2}{(141.4)^2} = 1$
Description of Weaker Responses	*Weaker responses* showed limited understanding of ellipse terminology and the structure of its standard equation. Some candidates confused the maximum and minimum distances, leading to incorrect calculations of the semi-major axis and focus. Others incorrectly assumed the centre was at the origin or failed to relate the given distances to the geometry of the ellipse. Several responses lacked the correct formula or used the wrong values in forming the equation, resulting in inaccurate or incomplete solutions.
Images of Weaker Responses	**Image (i)** b = 129.5 b^2 = 16770.25 c = 84.077 a = 154.4 a^2 = 23839.36 So $b^2 = a^2 - c^2$ foci (±84.077, 0) $c^2 = a^2 - b^2$ = 23839.36 - 16770.25 c^2 = 7069.11 ii. equation representing the orbit of Mars: $\frac{x^2}{23839.36} + \frac{y^2}{16770.25} = 1$ a^2 = 23839.36, b = 16770.25

	Image (ii) 

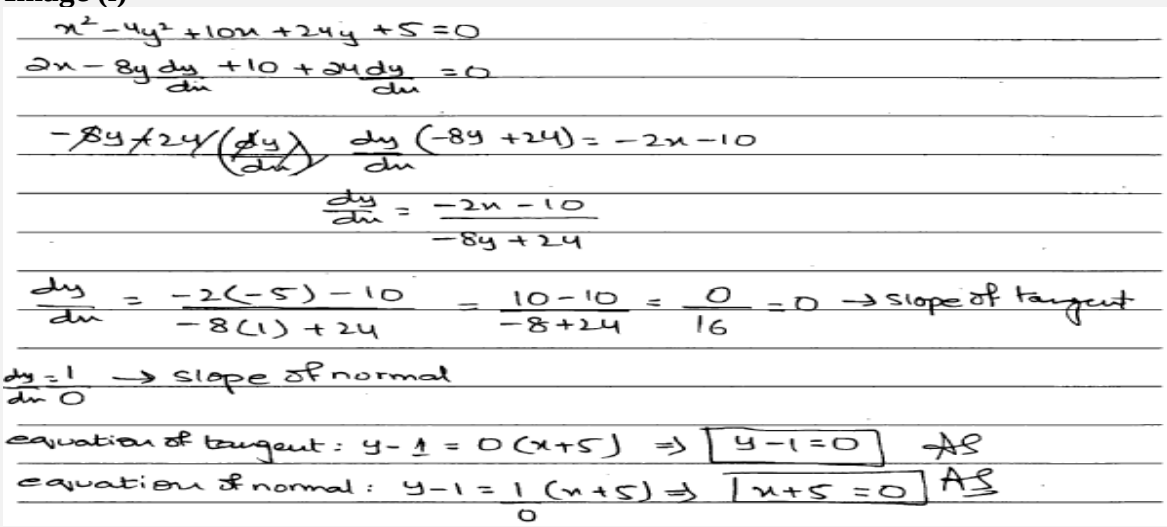
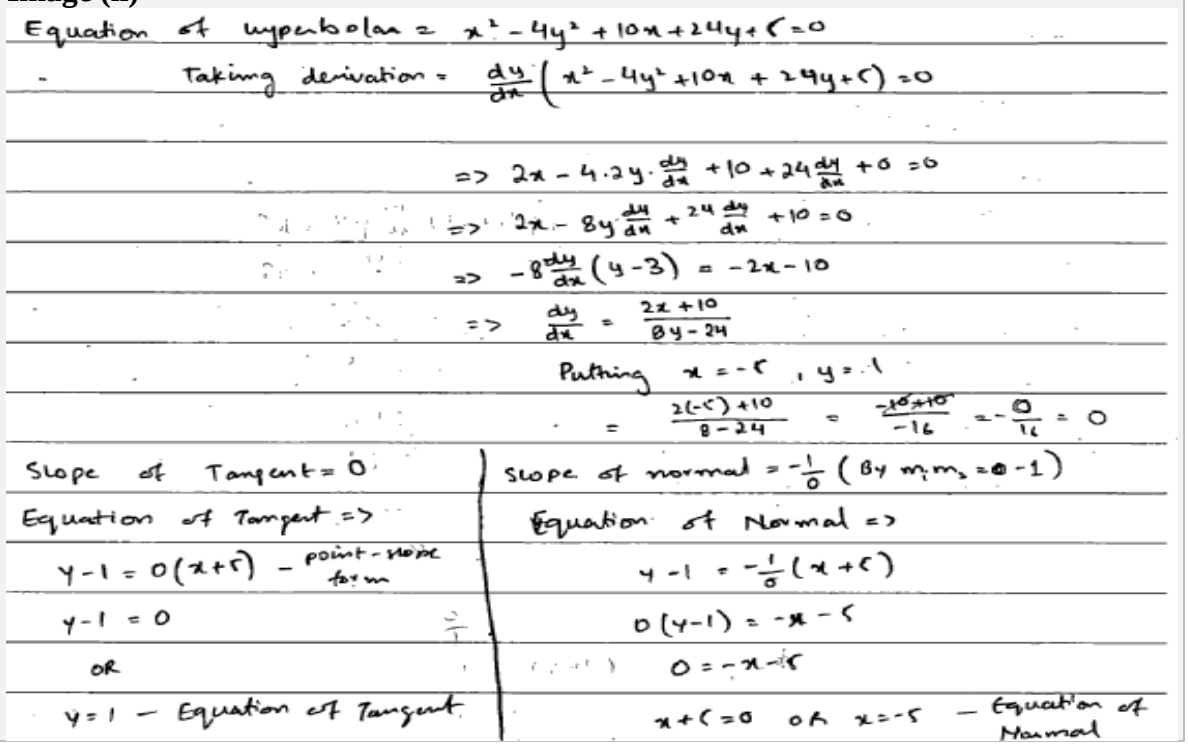
Suggestions for improvement (Highlight all that apply)

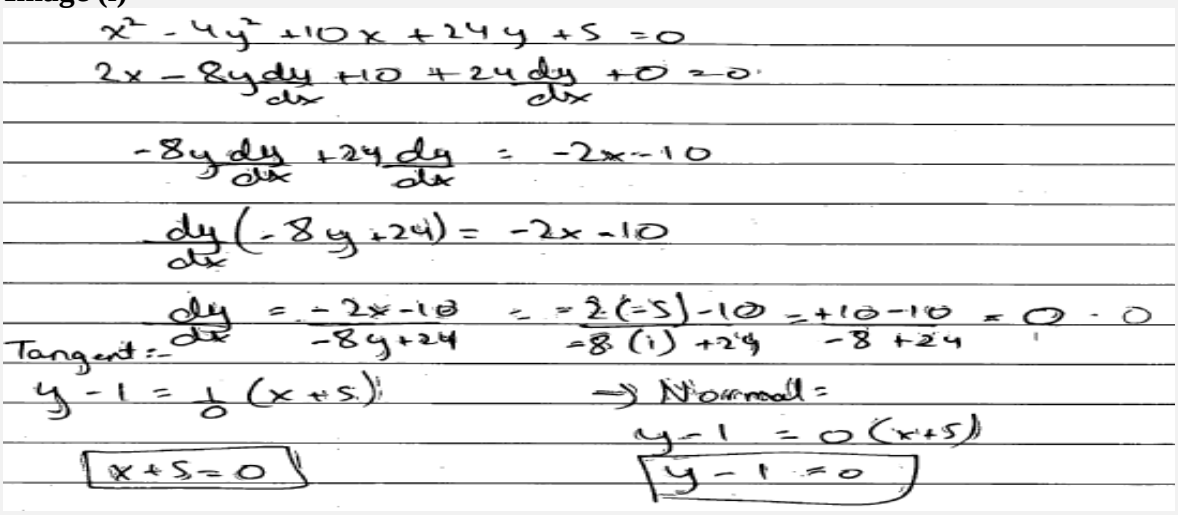
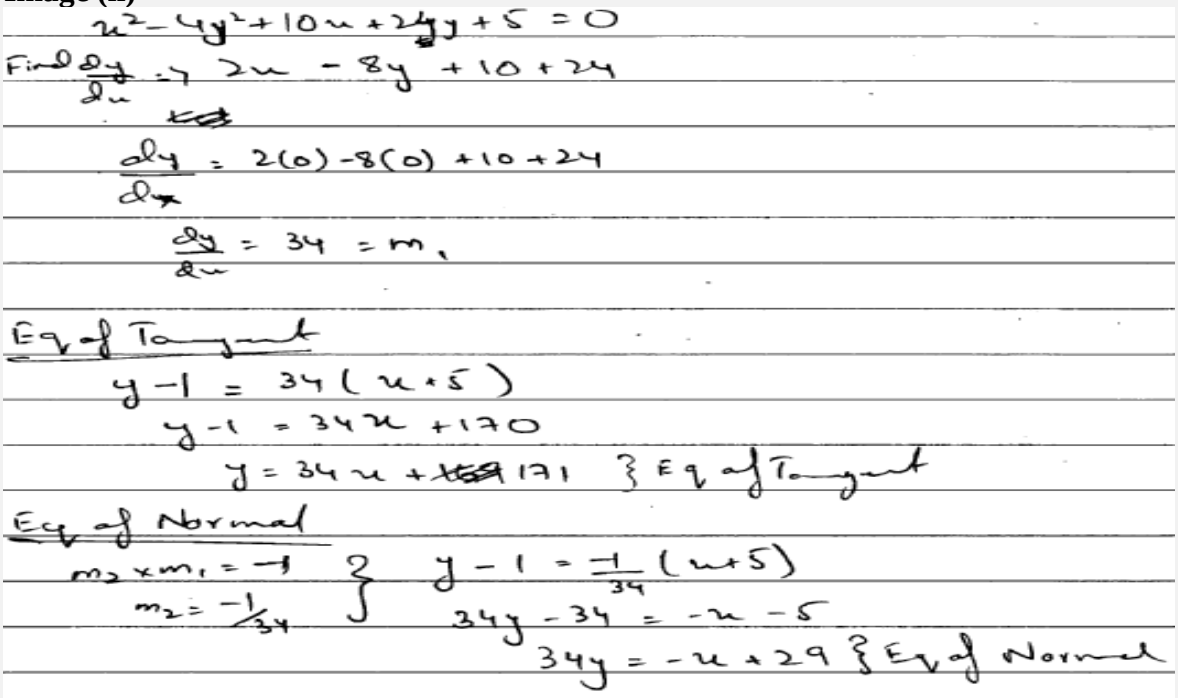
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Any Additional Suggestion:

- Use GeoGebra to plot ellipses from given equations and label centre, vertices, co-vertices, and foci.
- Give a matching game where one card shows an ellipse equation and another shows its key parameters.
- Provide incomplete steps for finding the centre, axes lengths, and eccentricity; students fill in the missing calculations.
- Organise a relay where one student converts the general equation to standard form, another identifies elements, and the third solves a related word problem.
- Give real-life contexts (planetary orbits, whispering galleries) and have students model them as ellipses and solve for required distances.

Question No. 7c Candidates were given the choice to attempt any TWO out of the three questions: 7a, 7b and 7c.	
Question Text	Find the equations of tangent and normal to the hyperbola $x^2 - 4y^2 + 10x + 24y + 5 = 0$ at point (– 5, 1).
SLO No.	22.3.2
SLO Text	Find the equation of tangent and normal to a hyperbola: a. at a point b. which is parallel to a given line c. which is perpendicular to a given line.
Max Marks	4

Cognitive Level	A
Checking Hints	1 Mark for slope of tangent at (–5,1) 1 Mark for equation of tangent 1 Mark for slope of normal 1 Mark for equation of normal
Overall Performance	Overall, a good number of candidates attempted this question, with many achieving full marks and others providing partially correct answers. The responses reflected a range of understanding, while some candidates demonstrated a strong grasp of the concept, others struggled to recall and correctly apply the formulae for tangent and normal.
Description of Better Responses	Candidates who performed well demonstrated a strong understanding of the properties and equations related to hyperbolas. They correctly substituted the given point into the general form to determine if it lay on the curve. These candidates then successfully differentiated the equation implicitly (or used standard results), found the slope at the point, and applied the point-slope form to obtain the equation of the tangent. For the normal, they used the concept of perpendicular slopes accurately and applied it to the same point. Their work was mathematically sound, logically structured, and free from errors.
Images of Better Responses	**Image (i)**  $x^2-4y^2+10x+24y+5=0$ $2x-8y\frac{dy}{dx}+10+24\frac{dy}{dx}=0$ $-8y+24\left(\frac{dy}{dx}\right)\ \frac{dy}{dx}(-8y+24)=-2x-10$ $\frac{dy}{dx}=\frac{-2x-10}{-8y+24}$ $\frac{dy}{dx}=\frac{-2(-5)-10}{-8(1)+24}=\frac{10-10}{-8+24}=\frac{0}{16}=0 \rightarrow$ slope of tangent $\frac{dy}{dx}=\frac{1}{0} \rightarrow$ slope of normal equation of tangent: $y-1=0(x+5) \Rightarrow \boxed{y-1=0}$ AS equation of normal: $y-1=\frac{1}{0}(x+5) \Rightarrow \boxed{x+5=0}$ AS **Image (ii)**  Equation of hyperbolas = $x^2-4y^2+10x+24y+5=0$ Taking derivation = $\frac{dy}{dx}(x^2-4y^2+10x+24y+5)=0$ $\Rightarrow 2x-4.2y.\frac{dy}{dx}+10+24\frac{dy}{dx}+0=0$ $\Rightarrow 2x-8y\frac{dy}{dx}+24\frac{dy}{dx}+10=0$ $\Rightarrow -8\frac{dy}{dx}(y-3)=-2x-10$ $\Rightarrow \frac{dy}{dx}=\frac{2x+10}{8y-24}$ Putting $x=-5$, $y=1$ $=\frac{2(-5)+10}{8-24}=\frac{-10+10}{-16}=-\frac{0}{16}=0$ Slope of Tangent = 0 Equation of Tangent => $y-1=0(x+5)$ – point-slope form $y-1=0$ OR $y=1$ – Equation of Tangent Slope of normal = $-\frac{1}{0}$ (By $m_1m_2=-1$) Equation of Normal => $y-1=-\frac{1}{0}(x+5)$ $0(y-1)=-x-5$ $0=-x-5$ $x+5=0$ OR $x=-5$ – Equation of Normal

Description of Weaker Responses	*Weaker responses* revealed a lack of clarity in applying differentiation or in using the standard tangent and normal formulae for hyperbolas. Some candidates confused the roles of tangent and normal or applied incorrect slope relationships. Errors were also seen in forming the equations after finding the slope, such as incorrect substitution or misuse of formulae. These mistakes indicate the need for more focused practice in both the calculus and coordinate geometry aspects of conic sections.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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• Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:

- Use GeoGebra to plot a hyperbola and a chosen point on it, then draw and verify the tangent and the normal at that point.
- Give a card set with points on the hyperbola; students compute and check tangent and normal at each point against the graphing tool.
- Provide a target slope (parallel to a given line); students find all hyperbola points where the tangent has that slope and verify visually.
- Run a "parallel hunt" relay: one student finds the required slope from the given line, another finds candidate points, a third writes and checks the tangent equation on the graph.
- Provide a line and ask for tangents perpendicular to it; students determine the negative reciprocal slope, locate the point(s) on the hyperbola, and confirm with a plotted normal.
- Quick-fire challenge: teacher flashes a graph with a drawn line; teams shout "parallel" or "perpendicular," then sketch the corresponding tangent or normal and validate on Desmos.
- Error-spotting sheets: show near-correct tangents/normals for each case; students identify the mistake and correct it using the graphing tool as evidence.

Question No. 8	
Question Text	Find the transformed equation of $4x^2 - 4y = 0$ with respect to the new axes obtained by rotation of axes about the origin through an angle of $60°$.
SLO No.	23.1.3
SLO Text	Find the transformed equation by using translation or rotation of axes.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for writing the equation of transformation 1 mark for application of two transformation equations 1 mark for substituting the values of x and y in the equation 1 mark for finding the required equation of transformation
Overall Performance	Overall, candidates generally found this question difficult. A few showed understanding of the concept and used the correct method for rotating axes, but most struggled with the transformation process. The question required accurate use of coordinate rotation formulae, which many candidates were unable to recall or apply correctly.
Description of Better Responses	Candidates who performed well demonstrated a clear understanding of the rotation of coordinate axes. They correctly used the rotation formulae and substituted the expressions into the original equation. They accurately used the given angle of 60°, computed the trigonometric values, and substituted these into the transformation. These candidates simplified the resulting expression correctly, arriving at the transformed equation without error. Their responses were logically presented and mathematically consistent.

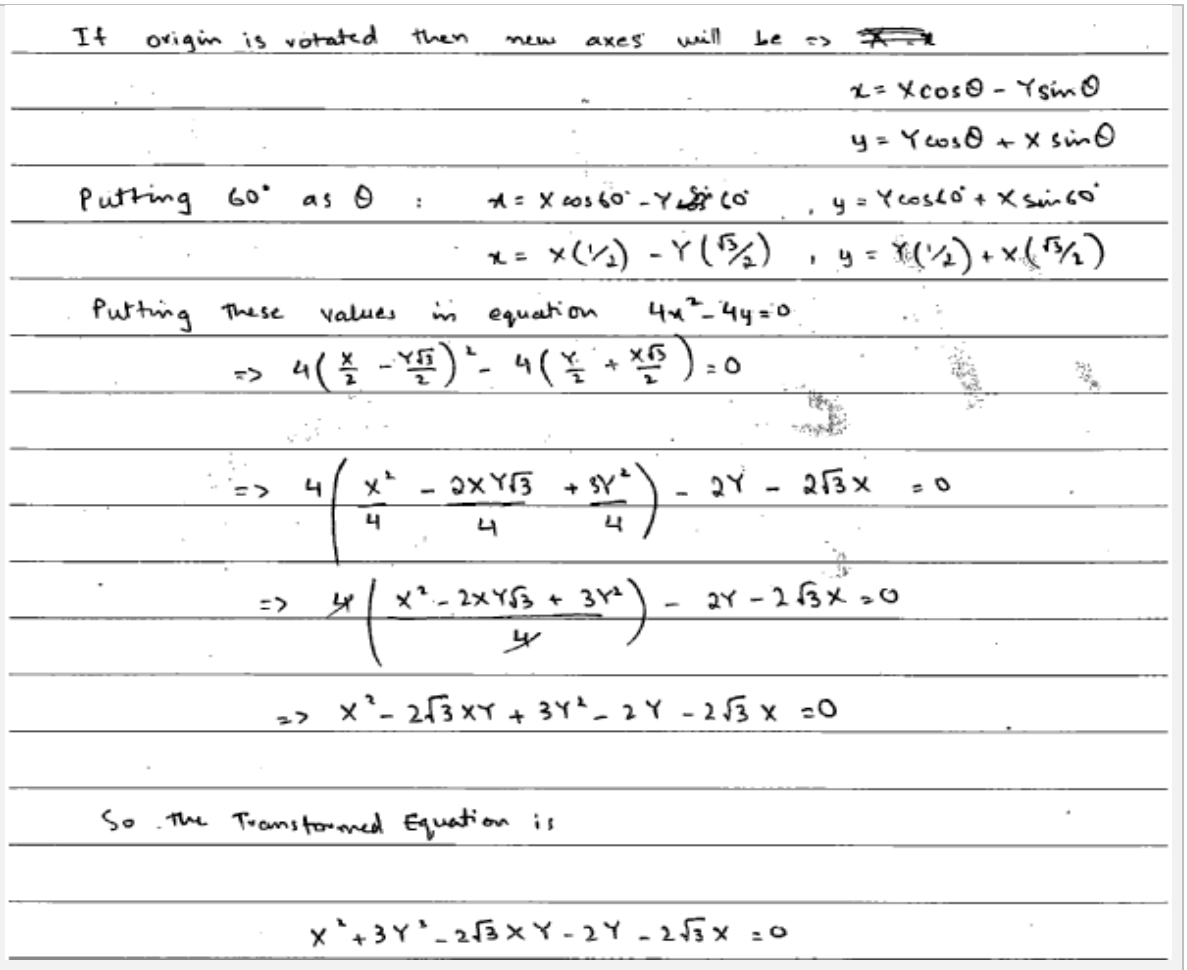
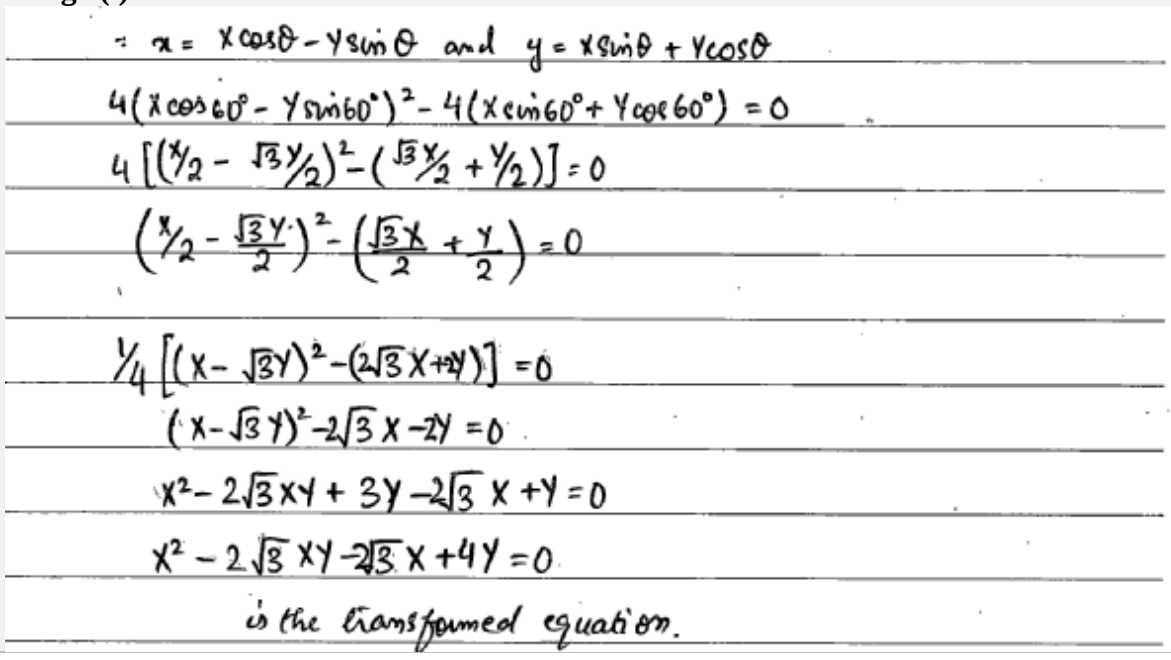
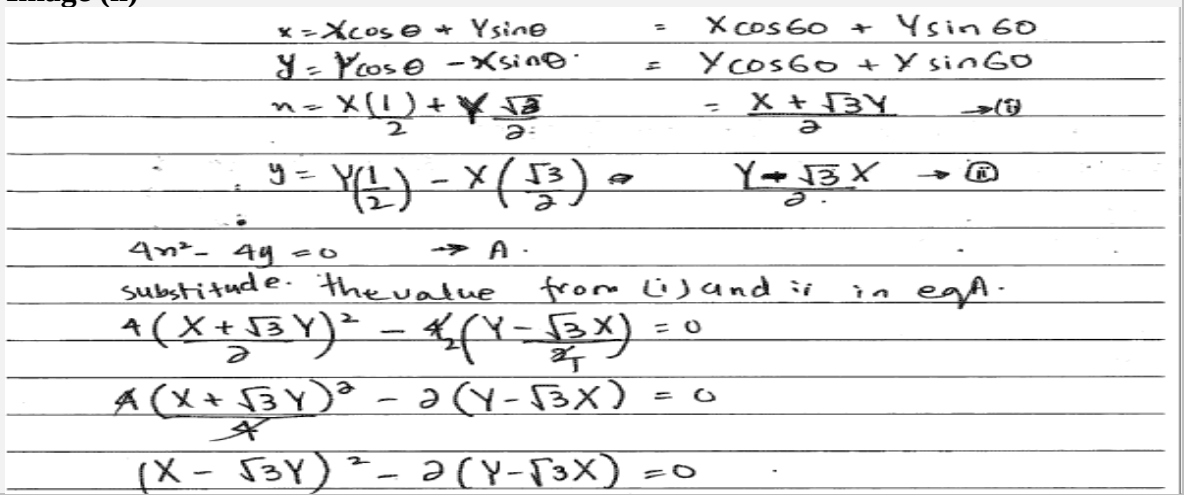
Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed a lack of familiarity with the correct rotation formulae. A common mistake was the incorrect use of transformation equations, $X = x\cos\theta + y\sin\theta$, and $Y = y\cos\theta - x\sin\theta$, which are not suitable for this context. As a result, when they substituted the angle of 60°, the expressions for x and y were incorrect, leading to the wrong transformed equation. In other cases, even when the correct expressions were used, several candidates made algebraic errors or failed to simplify the resulting equation fully. These issues reflected gaps in understanding both the geometric concept of rotation and the algebraic manipulation needed to complete the process.
Images of Weaker Responses	**Image (i)** 

Image (ii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Give a "before–after" matching set: original graphs on one side, transformed equations on the other; students pair them.
- Run a relay: one student translates coordinates, the next rotates them, the last writes the final equation and checks on a graphing tool.
- Provide mixed transformations; students decide the order (translate then rotate or rotate then translate) and verify which matches the target graph.
- Error-spotting sheets: near-correct transformed equations with small mistakes; students find and fix them using overlays in Desmos.
- Challenge students to recover the original equation by applying the inverse transformation to a given transformed equation.
- Station rotation: at each station, students perform a specific transform (right/left shift, up/down shift, rotation by a fixed angle) and stamp the updated equation.
- Real-world alignment: give an image of a tilted object (e.g., road sign); students model it with a rotation of axes and confirm the simplified equation.
- "Guess the transform" game: display two graphs; teams state the translation and rotation that maps one to the other, then write the transformed equation.
- Composition puzzle: students are given small "transform cards" (shift right 3, shift up 2, rotate by 45 degrees) and must compose them to reach a target equation, verifying on a graph.

Question No. 9a

Candidates were given the choice to attempt any ONE out of the two questions: 9a and 9b.

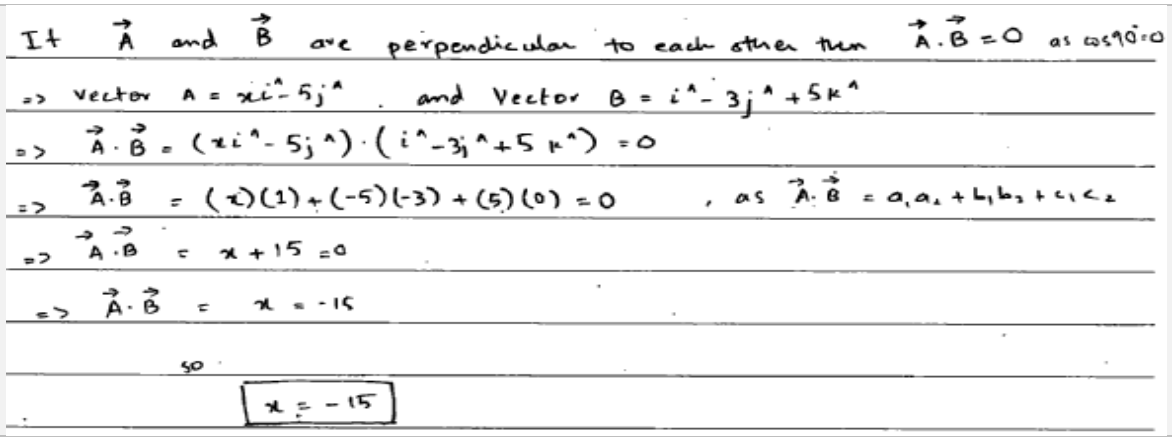
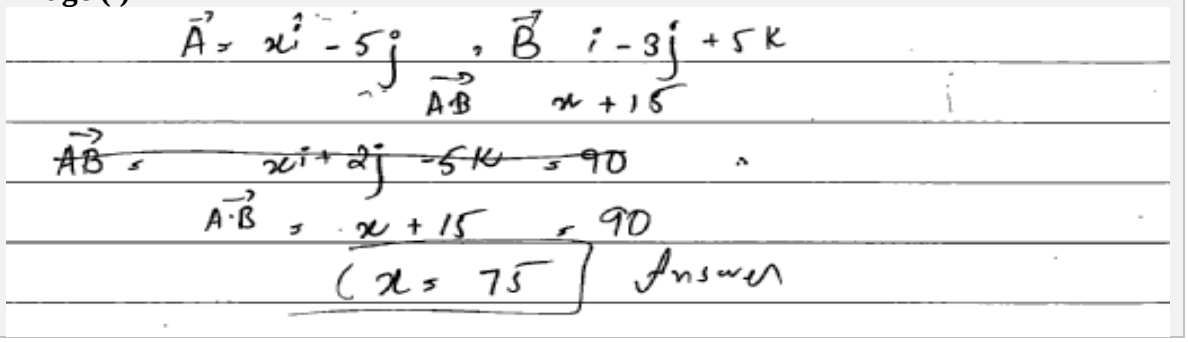
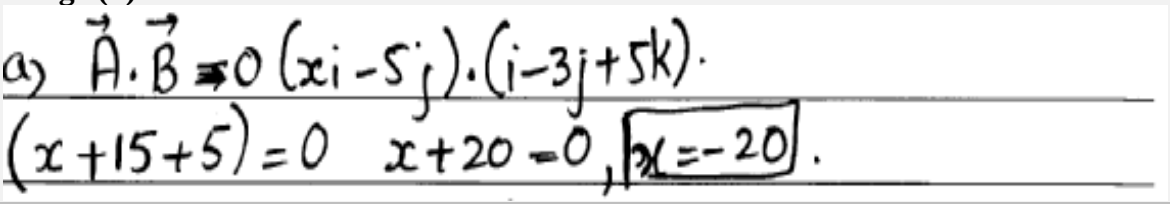
Question Text	Find the value of x for which the two vectors $\vec{A} = x\hat{i} - 5\hat{j}$ and $\vec{B} = \hat{i} - 3\hat{j} + 5\hat{k}$ are perpendicular to each other.
SLO No.	24.3.5 (2.3.4)
SLO Text	Solve the problems based on the condition of orthogonality of two vectors. (Find dot product of two vectors)
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for calculation of $\vec{A}.\vec{B}$ 1 mark for applying condition of perpendicular vectors as $\vec{A}.\vec{B} = 0$ 1 mark for value of x
Overall Performance	The majority of candidates attempted this part, but only some answered it successfully, demonstrating a good understanding of the scalar (dot) product and correctly applying the property of the dot product to find the value of x. Others, however, struggled with the concept and made errors either in calculating the dot product or in solving the resulting equation.
Description of Better Responses	Candidates who performed well showed a strong grasp of the scalar product concept. They accurately applied the formula for the dot product of two vectors and carefully substituted the given components. These candidates then solved the resulting algebraic equation with precision, correctly isolating the variable x. Their responses were structured, showing each step of the calculation logically, and concluded with the correct answer without any mathematical errors.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed difficulty in calculating the scalar product. Some candidates incorrectly multiplied vector components, while others failed to sum the products correctly. A few of the candidates misapplied the scalar product definition or attempted irrelevant vector operations, indicating gaps in conceptual understanding. These errors often resulted in an incorrect or incomplete solution, reflecting a need for more practice in basic vector operations and equation solving.
Images of Weaker Responses	**Image (i)** 

	Image (ii) 

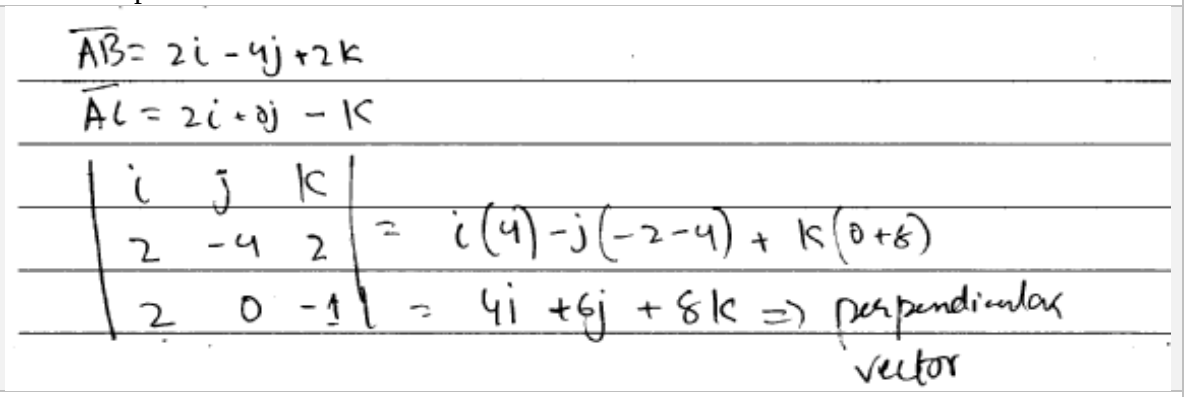
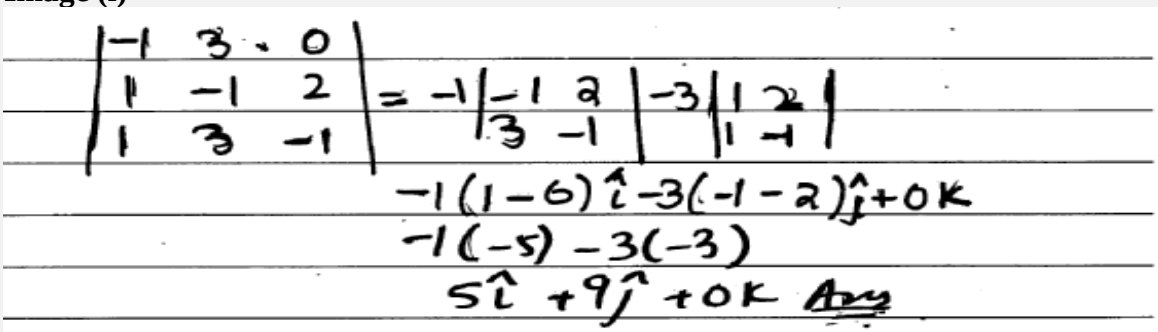
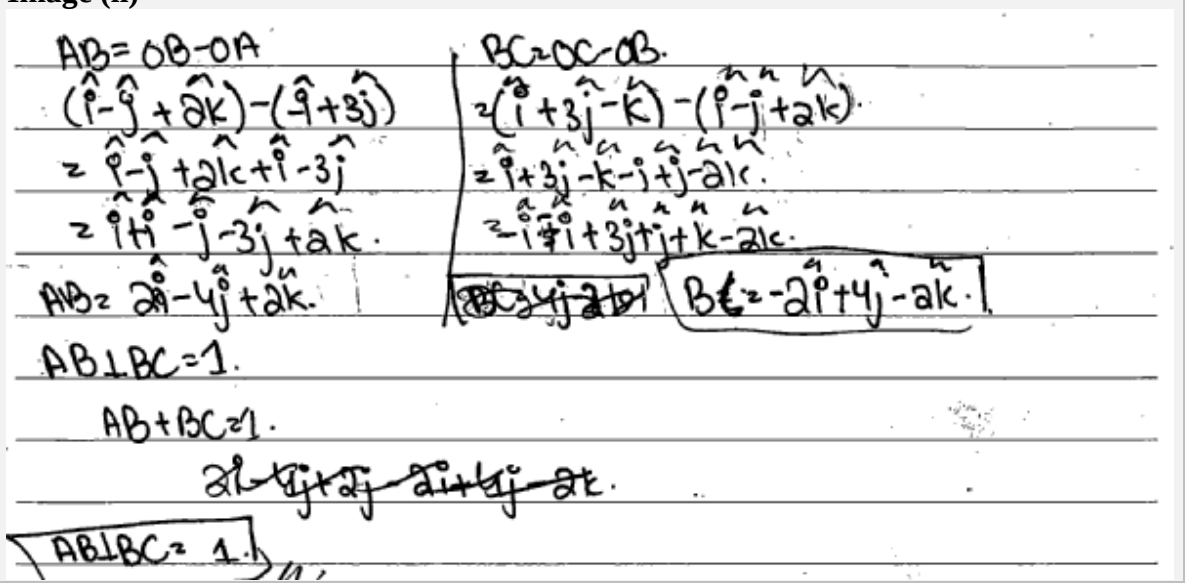
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

- Use GeoGebra 3D to plot two vectors and visually check if they are perpendicular.
- Match vector pairs with their dot product values, marking which are orthogonal.
- In pairs, students race to compute dot products and decide if vectors are perpendicular.
- Provide incomplete dot product calculations for students to finish and verify.
- Real-life link: model forces or directions as vectors and check orthogonality using the dot product.

Question No. 9b Candidates were given the choice to attempt any ONE out of the two questions: 9a and 9b.	
Question Text	Find a vector perpendicular to the plane containing points $A(-1, 3, 0)$, $B(1, -1, 2)$ and $C(1, 3, -1)$.
SLO No.	24.4.10
SLO Text	Solve problems related to SLOs 24.4.4 and 24.4.9. (24.4.4. find cross product of two vectors) (find angle between two vectors using cross product.)
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for finding two vectors AB and AC form given points 1 mark for applying cross product for finding perpendicular vector to the plane 1 mark for finding a perpendicular vector to the plane
Overall Performance	Overall, a smaller number of candidates attempted this question, and many of them gave only partially correct responses. Those with a clear understanding of the cross product and the relevant formulae were able to work out the vector cross product correctly. However, others struggled to recall and apply these concepts accurately, which affected the correctness of their answers.

Description of Better Responses	Candidates who performed well clearly understood how to form vectors such as *AB*, *BC*, or *AC* from given points. They correctly calculated the direction vectors by subtracting position vectors and then used the determinant method to compute the cross product. These candidates demonstrated accurate handling of the calculation, correctly determining both the magnitude and direction of the resulting vector. Their work was structured, and they understood that the cross product represents a vector perpendicular to the plane formed by the two input vectors.
Image of Better Response	
Description of Weaker Responses	Many of the *weaker responses* showed limited understanding of the cross product. A number of candidates were unable to correctly find vectors such as *AB*, *BC*, or *AC*, resulting in incorrect direction vectors. Others confused the cross product with the scalar (dot) product, using inappropriate methods. Some attempted to apply the correct cross product formula but made errors in calculating determinants or signs. Additionally, several candidates were unsure about the geometric interpretation of the cross product and failed to identify the resulting perpendicular vector. These issues indicate a need for greater familiarity with both the conceptual meaning and mechanical steps involved in solving cross product problems.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

- Use GeoGebra 3D to plot two vectors and their cross product vector, noting its perpendicular direction.
- Match vector pairs with the magnitude of their cross product.
- In pairs, students calculate cross products and verify by checking perpendicularity.
- Provide partially completed cross product determinants for students to finish.
- Use the formula $|A \times B| = |A||B|\sin\theta$, to find angles, then confirm with a graphing tool.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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